Unexpected Online Financial News and Intraday Market Reactions: An Event-Study of EUR/USD and Nasdaq-100 Prices

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Abstract

We study the intraday response of EUR/USD and the Nasdaq-100 index to 2,449 unscheduled news events from a public real-time newsfeed over a thirteen-month sample. Each event is independently labelled by a large language model and benchmarked against the Loughran-McDonald dictionary and FinBERT-tone. Using a matched, hour-of-day and day-of-week stratified baseline with cluster-robust standard errors and uniform Benjamini-Hochberg false discovery rate adjustment, we find that news events are associated with intraday magnitude 1.3× to 1.9× the baseline level on every asset-window cell. The directional content of LLM sentiment is modest (49–54% hit rate) and below transaction costs in a backtest, but the LLM outperforms both baselines on direction by 2–6 percentage points and produces asset-specific sentiment on 73.5% of events. On the Financial PhraseBank external benchmark, the LLM attains 84.6% accuracy and Cohen's $\kappa = 0.72$ against gold human labels, outperforming FinBERT by 5.4 percentage points. The pre-event absolute return matches the post-event return in magnitude, indicating that the public headline timestamp is not the informational event time. The full pipeline is released with seeded reproducibility.

Keywords: event study, intraday returns, sentiment analysis, large language models, financial dictionaries, news, EUR/USD, Nasdaq-100, false discovery rate.

JEL classification: G14, G17, C45, C58.

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1 Introduction

Information enters financial markets through a heterogeneous network of public news services, professional terminals, broker desks, and, in the last decade, real-time social and chat-based aggregators that an increasing share of retail and small-shop traders monitor as their primary news feed. These newer channels carry the same headlines that appear on Bloomberg or Reuters but at a variable latency and with a publication timestamp that is not the timestamp of the underlying primary source. Whether and how prices react to the arrival of news on such channels—and whether the directional content of those headlines can be extracted in a useful way by modern language models—is a question with direct practical relevance to a large and growing audience of intraday traders, and a methodological question for the event-study literature that has historically focused either on scheduled macro releases (Andersen et al. 2003) or on slower-moving newspaper text (Tetlock 2007; Loughran and McDonald 2011).

This paper studies the relation between unscheduled news headlines from a representative real-time news service—the FinancialJuice Discord newsfeed—and intraday price reactions in the EUR/USD spot exchange rate and the Nasdaq-100 index, over a thirteen-month sample from 24 March 2025 to 5 May 2026. The dataset comprises 2,449 hand-filtered "gold" news events drawn from 63,016 raw messages and aligned to one-minute price bars on both assets. Each event is independently labelled by a large language model with a discrete sentiment for USD and for NDX, an expected magnitude, a surprise level, and a self-reported confidence. We test fourteen pre-specified hypotheses covering volatility, direction, persistence, time-of-day, cross-asset behaviour, and pre-event drift, and we report the central findings with Benjamini-Hochberg false discovery rate adjustment (Benjamini and Hochberg 1995) applied uniformly across the full hypothesis battery.

Three findings warrant emphasis. First, the news events are robustly associated with elevated intraday magnitude: the mean absolute return, the high-low range, and the maximum absolute intra-window move are between 1.3× and 1.9× their matched-baseline counterparts across all assets and windows tested, and the result is robust to winsorisation at the 1% tails, to the introduction of multivariate controls for category, surprise, magnitude, confidence, cluster size, and pre-event drift, and to a pre/post split at the LLM's training data cutoff. The intra-window range and maximum-absolute-move statistics are particularly robust, with q-values below 10^{-30} on every cell. Second, the directional content of LLM-derived sentiment is modest: direction hit rates range from 49% to 54% on the primary windows, with only a small number of cells surviving the FDR correction. A backtest with realistic spread and slippage costs confirms that this edge is too small to be economically exploitable in the naive forms tested. Third, the LLM sentiment nonetheless out-performs both the standard Loughran-McDonald finance dictionary (Loughran and McDonald 2011) and the FinBERT-tone neural classifier (Yang et al. 2020) on direction hit rate by 2–6 percentage points across the primary windows, and produces cross-asset disagreement (different sentiment for USD and for NDX on the same event) on 73.5% of events—a property that neither single-polarity baseline can express by construction.

We also report two methodologically substantive findings. The absolute return in the fifteen minutes preceding a Discord headline is elevated above the matched baseline by a similar magnitude as in the post-event window. Without primary-source timestamps we cannot distinguish whether this reflects information arrival latency in the public feed or genuine pre-disclosure activity, but the practical implication for users of the feed is the same: entering a trade at the timestamp of the public headline is, on average, already late. And the sign of the +15-minute return agrees with the sign of the +4-hour return on 57.6% of events, with median absolute moves 3–4.5× larger over the longer window, so the initial reaction tends to be extended rather than reversed.

Our contribution is fourfold. We assemble a sizeable intraday event-study dataset on an actively-used public news feed and document its construction in enough detail to support replication. We propose an event-study workflow with matched-baseline standardisation, explicit cluster dependence handling, and uniform FDR adjustment as applied to a large battery of hypotheses. We compare an LLM-based sentiment classifier to two reference baselines—the Loughran-McDonald finance dictionary and the FinBERT-tone neural classifier—on identical event windows, providing direct quantitative support for the LLM methodology against both classical and BERT-era baselines. And we make the entire pipeline—including the validation harness, the LLM cache, the dictionary baseline, the FinBERT baseline, and the three-way comparison—public in a single repository with seeded reproducibility (see Reproducibility Statement).

The remainder of the paper is organised as follows. Section 2 reviews the relevant literature on event studies, financial sentiment analysis, and online news. Section 4 describes the data. Section 5 develops the methodology. Section 6 presents the results, grouped by theme. Section 7 discusses the implications. Section 8 details the limitations of the design and the directions for future work, and Section 9 concludes.

2 Literature Review

Our work intersects four established literatures. We organise the review around those four threads, taking care to identify, for each, the specific result that motivates a methodological choice in our own design or that frames the contribution of our findings.

2.1 Event studies and intraday price discovery

The event-study method, formalised in its modern shape by MacKinlay (1997), exploits the proposition that, under semi-strong market efficiency (Fama 1970), the price response to an informational event reveals its market-relevant content. The classical approach computes abnormal returns relative to a market model on daily data; the high-frequency extension reduces the observation horizon to intraday windows of seconds or minutes.

The canonical high-frequency study on macro news is Andersen et al. (2003), which uses six years of real-time foreign-exchange quotes to characterise the conditional mean response of major USD pairs to scheduled macroeconomic announcements. Two features of that literature inform our design. First, the appropriate benchmark in the high-frequency setting is a matched non-event window of the same length, not a long-horizon market model, because intraday volatility is dominated by hour-of-day and day-of-week seasonality that would otherwise contaminate the abnormal-return estimate. Second, on FX, USD strength is a derived quantity: a study that uses the EUR/USD price directly to test a USD-oriented hypothesis (e.g., "good news for the dollar produces a positive reaction") will assign the wrong sign to every event whose target asset is the dollar rather than the euro. We adopt both conventions: a matched, hour-of-day and day-of-week stratified baseline (Section 5.4) and an explicit USD proxy convention $r^{\mathrm{USD}} = -r^{\mathrm{EURUSD}}$ (Section 5.2).

Most of the existing high-frequency event-study literature focuses on scheduled macro releases—CPI, non-farm payrolls, central-bank decisions—because the release time is known in advance and clean event windows are straightforward to construct. Unscheduled news, such as geopolitical escalations or political statements, has received comparatively less attention at the intraday horizon. Our sample is dominated by such unscheduled events (we deliberately filter out the publisher's `$MACRO` tag, which marks scheduled releases), placing the analysis closer to the spirit of the news-text literature than to the macro-announcement tradition.

2.2 Financial sentiment analysis: from dictionaries to LLMs

The textual analysis of financial news as a predictor of returns begins with Tetlock (2007), who shows that the fraction of negatively-valenced words in the Wall Street Journal "Abreast of the Market" column predicts next-day index returns and reverses over the following week. Tetlock uses the Harvard IV-4 General Inquirer dictionary, an inheritance from political-science content analysis.

Loughran and McDonald (2011) subsequently establish that general-purpose sentiment dictionaries misclassify a large share of words in financial text—the canonical example is "liability," which is financially neutral but Harvard-negative—and they construct a finance-specific dictionary that has since become the standard baseline in the field. We use the Loughran and McDonald (2011) dictionary explicitly as our reference baseline against which the LLM-derived sentiment must improve in order to justify the methodological choice of an LLM (Section 6.2). We do not use Harvard IV-4, partly because Loughran and McDonald (2011) themselves demonstrate its inadequacy in this context.

Antweiler and Frank (2004) and Bollen et al. (2011) extend the textual-analysis program from professional newspaper text to user-generated content (stock message boards and Twitter respectively), demonstrating in both settings that simple bullishness counts have non-trivial predictive content for return and volatility, albeit small. Our study is in this tradition in that the FinancialJuice Discord feed is a real-time professional-curated aggregator broadcast to a primarily retail audience, rather than a peer-reviewed wire service.

The modern wave of work uses neural language models. Heston and Sinha (2017) benchmark a Thomson-Reuters proprietary neural network against Harvard and Loughran-McDonald on a large news corpus and find that the neural model captures both faster and more persistent reactions, particularly for negative stories. Yang et al. (2020) release FinBERT, a BERT-family model pre-trained on a 4.9-billion-token corpus of financial communications, with state-of-the-art results on the Financial PhraseBank benchmark. Calomiris and Mamaysky (2019) combine topic modelling with sentiment and unusualness ("entropy") signals on a multi-country Reuters corpus and show predictive content for country-level returns and volatilities at horizons up to one year. We extend this line of work to a contemporary large-language-model classifier (`deepseek-v4-flash`) applied to a real-time chat-aggregated news feed, with explicit benchmarking against the dictionary baseline and a self-consistency check against a more capable sibling model.

2.3 Online news, sentiment, and intraday reactions

The intersection of social or online news with intraday market behaviour has been studied at the daily horizon by Bollen et al. (2011) (mood from Twitter predicts the Dow Jones at several days' lead) and at the search-volume level by Da et al. (2015) (aggregated Google Trends queries for words like "recession" and "bankruptcy" predict short-term return reversals and volatility increases). Antweiler and Frank (2004) show that the volume rather than the direction of message-board posts predicts realised volatility—a result we revisit in Section 6.6 with our log(cluster size) control, where larger news bursts produce larger magnitude reactions even after controlling for the LLM sentiment and category labels.

The relatively under-explored corner of this map is the sub-fifteen-minute response to unscheduled news on a real-time public chat feed, which is precisely the setting our dataset addresses.

2.4 Multiple testing and reproducibility

A practical concern when running a large pre-specified hypothesis battery is that the number of tested cells inflates the family-wise type-I error far beyond the nominal level. We follow the Benjamini-Hochberg procedure (Benjamini and Hochberg 1995) and report q-values that control the false discovery rate across the union of all p-values produced by the pipeline. The procedure is more appropriate than the Bonferroni correction for a large hypothesis family in which a non-trivial fraction of nulls are expected to be false. All conclusions in Section 6 are drawn from q-values, with raw p-values reported for transparency. The full pipeline, the seed, and the LLM cache are documented in the Reproducibility Statement to support direct replication.

3 Hypothesis development

The empirical analysis tests eight pre-specified main hypotheses (H1–H14, restricted to the eight with non-trivial findings) and seven post-specification extensions (C1–C7). This section motivates the eight main hypotheses from theory and prior evidence; the full list with formal test specifications appears in Section 5.5.

3.1 Magnitude response (H1)

Under semi-strong market efficiency (Fama 1970), the arrival of an unscheduled informational event should produce a price revision that is detectable in the conditional second moment of returns even when its conditional mean is small or sign-ambiguous. Andersen et al. (2003) demonstrate this directly for scheduled macro announcements in high-frequency FX, and Calomiris and Mamaysky (2019) document analogous volatility effects for unscheduled news in equity markets cross-country. We hypothesise that, on intraday windows from one to two hundred and forty minutes, the absolute event-window return exceeds the absolute matched-baseline return after controlling for hour-of-day and day-of-week seasonality. This is H1.

3.2 Directional content of sentiment labels (H2)

The textual-analysis tradition since Tetlock (2007) documents that sentiment extracted from news text predicts future returns, though the directional signal is small in absolute terms. Heston and Sinha (2017) report that neural sentiment classifiers extract richer directional content than dictionary classifiers (Loughran and McDonald 2011), particularly for negative news. We hypothesise that the discrete bull/bear LLM sentiment predicts the realised target direction at hit rates above 50%, with effect sizes in the single-digit percentage points consistent with prior literature. This is H2.

3.3 Sentiment-trend interaction (H3)

The classical reversal literature (Tetlock 2007) suggests that the directional content of sentiment depends on prior price behaviour: bullish news following a price drop produces a larger upward revision than bullish news following a price run-up, because the latter has been at least partially priced in. We hypothesise that the interaction between LLM sentiment and the prior 60-minute trend is a significant predictor of the absolute event-window return, controlling for the sentiment and trend main effects. This is H3.

3.4 Closed-period gap (H4)

When markets are closed (weekends, holidays), news that arrives during the closure should be impounded into the opening price discontinuously rather than continuously. The cumulative target sentiment across the closure should therefore predict the realised opening-bar gap, in line with Fama (1970) and standard event-study results on overnight returns. This is H4.

3.5 Pre-event drift and information arrival timing (H8)

The textual-analysis literature typically assumes that the public publication time of a news headline coincides with the informational event time, which is appropriate for machine-readable wire feeds with millisecond timestamps. For real-time chat-aggregated public feeds, the latency between the primary source and the public publication is variable and can be substantial. We hypothesise that the absolute return in the fifteen-minute window preceding the public timestamp exceeds the matched baseline at a magnitude comparable to the post-event drift, consistent with primary-source latency rather than with insider trading or systematic front-running. This is H8.

3.6 Persistence (H9)

If the news effect represents a permanent re-pricing rather than a transient liquidity shock, the sign of the short-horizon return should agree with the sign of the longer-horizon return at a rate above 50%. The classical asset-pricing literature (Fama 1970; MacKinlay 1997) does not distinguish these mechanisms at intraday frequencies. We hypothesise that the sign of the +15-minute target return agrees with the sign of the +4-hour target return at a rate above 50%, with the magnitude of the longer-horizon return larger than the magnitude of the shorter-horizon return. This is H9.

3.7 Time-of-day heterogeneity (H11)

Intraday volatility is strongly heterogeneous across the trading day, with established peaks around major exchange openings and macro-release windows (Andersen et al. 2003). The event/baseline magnitude ratio should reflect this heterogeneity if our matched-baseline construction succeeds at absorbing the intraday cycle. We hypothesise that hour-of-day effects on the event/baseline ratio are statistically detectable for at least one of the two target assets. This is H11.

3.8 Cross-asset relation (H14)

Macro-news events typically affect USD and equity markets through correlated risk-on/risk-off mechanisms. Andersen et al. (2003) document positive correlation between USD strength and S&P 500 returns conditional on macro news. We hypothesise that, under the USD-proxy convention, the event-window USD return and the event-window NDX return are positively correlated. This is H14.

3.9 Summary

The eight main hypotheses cover the four dimensions on which an event-study of unscheduled online news should provide evidence: magnitude (H1), direction (H2 and H3), timing (H4, H8, H9), and cross-sectional structure (H11, H14). The remaining tests in the pipeline (H5–H7, H10, H12–H13 and C1–C8) are auxiliary or robustness extensions, motivated below where they are reported.

4 Data

4.1 News data

We use the public newsfeed of the FinancialJuice Discord server, captured via DiscordChatExporter into a single JSON archive. The export covers the period from 24 March 2025 to 5 May 2026 (approximately thirteen months) and contains 63,016 individual messages from the official `FinancialJuice` account. The feed is a real-time aggregation of macroeconomic releases, central bank communication, geopolitical headlines, political statements, energy market news, and corporate announcements, intended primarily for an audience of intraday traders. All timestamps are recorded by Discord at the moment the message is published and converted to UTC.

To isolate market-relevant unscheduled events from routine data releases, we apply a deterministic "gold" filter: a message is gold if (i) it is flagged with a red dot or begins with BREAKING, and (ii) it is not flagged with the `$MACRO` tag that the publisher attaches to scheduled macro indicator releases (CPI, NFP, PMI, etc.). The first condition selects headlines that the publisher itself considers high-impact; the second removes scheduled releases that would otherwise dominate the sample and whose timing carries no informational surprise. The filter retains 2,449 messages (3.9% of the raw feed), which we treat as the working population of unscheduled news events.

We assign each gold event to one of seven categories using deterministic keyword rules embedded in `parse_fj_discord.py`: `macro_release` (residual data releases not caught by the `$MACRO` tag), `central_bank` (Federal Reserve, ECB, Bank of England, etc.), `geopolitical` (war, sanctions, diplomacy), `politics` (executive and legislative statements), `energy` (OPEC, oil inventories, gas), `corporate` (named firm news), and `other`. The category labels are used only for stratified hypothesis testing in later sections and play no role in event selection.

4.2 Sentiment labels

Each gold event is independently labelled by a large language model with respect to its short-horizon directional implication for the U.S. Dollar (USD) and the Nasdaq-100 Index (NDX). We use `deepseek-v4-flash` accessed via an OpenAI-compatible API endpoint. The system prompt (reproduced in the Online Appendix) restricts the model to a strict JSON schema with the following fields per event i: a discrete USD sentiment $S_i^{\text{USD}} \in \{\text{bull}, \text{bear}, \text{neutral}\}$, a discrete NDX sentiment S_i^{NDX} in the same support, signed directional strengths in $[-1, +1]$ for each asset, an expected magnitude in $\{\text{low}, \text{med}, \text{high}\}$, a surprise level in $\{\text{expected}, \text{surprise}, \text{shock}\}$, a self-reported confidence in $[0, 1]$, and a one-sentence rationale. The prompt explicitly instructs the model to (a) base its judgment only on the supplied text, (b) return `neutral` with low confidence when the message is ambiguous, and (c) treat USD and NDX independently so that risk-off events can map to `bull USD, bear NDX` simultaneously. Five few-shot examples covering geopolitical, central bank, corporate, and neutral cases are included verbatim in the system prompt to anchor the output format. Calls are cached locally in SQLite, keyed on the hash of the model name, system prompt, and event content, so that re-runs during prompt iteration do not duplicate API costs.[1]

The LLM-derived labels are the primary sentiment input throughout the paper. In Section 6.2 we benchmark them against a deterministic Loughran-McDonald dictionary baseline (Loughran and McDonald 2011) and against the FinBERT-tone neural classifier (Yang et al. 2020).

[1]We set temperature to zero throughout and validate cache hits against fresh calls on a held-out subsample of 50 events; agreement is exact at this temperature with the API version used during data collection.

4.3 Price data

We use one-minute open-high-low-close-volume (OHLCV) bars for two target assets from Dukascopy: the EUR/USD spot exchange rate and the `E_NQ-100` Nasdaq-100 contract for difference (CFD). The CFD price track is used as a tradable proxy for the cash Nasdaq-100 index in the absence of an authoritative consolidated tape at one-minute resolution. We acknowledge the proxy nature of this series in Section 8 and discuss its implications for the reported volume tests.

The price coverage spans 24 March 2025 to 5 May 2026, identical to the news coverage window. After de-duplication and gap detection, the EUR/USD panel contains 416,500 bars and the NDX panel contains 381,122 bars; the difference reflects the venue's CFD market hours (no weekend trading, exchange holidays).[2] Closed-period gaps are detected automatically and events that fall inside them are diverted to the closed-period gap test (Section 6.5) rather than to the main intraday windows. A reported volume series accompanies each bar; for the NDX CFD the series is best understood as tick volume rather than consolidated exchange volume, and any volume-based test must be interpreted accordingly.

4.4 Final analysis sample

The two coverage windows (news and prices) are identical, so the intersection used for the event study is also 24 March 2025 to 5 May 2026, and no gold events are dropped for falling outside the price range. After cluster assignment at a 15-minute gap threshold, the 2,449 events organise into 1,405 distinct news clusters; the median cluster contains a single headline, but the largest contain dozens of related releases issued within minutes of each other (typically central bank decisions followed by press conferences, or geopolitical escalations followed by official reactions). Cluster identity is preserved throughout the analysis: non-regression tests use the first event per cluster as an independent observation, and regression standard errors are clustered on cluster identity.

Generating one event-asset-window record for each combination of (event, asset in {EUR/USD, NDX}, window in {1, 5, 15, 60, 240} minutes) yields a panel of 24,490 event-window rows. A parallel cluster-level panel constructed from the first timestamp of each cluster yields 14,050 rows. Both panels are written to the project repository (`event_study_windows.csv` and `cluster_event_study_windows.csv`). Table 1 reports the key sample quantities, Table 2 defines the variables used throughout the analysis, and Table 3 provides descriptive statistics for the key event-level and window-level quantities.

[2] Bar count discrepancies between EUR/USD and NDX of approximately 9% are consistent with the additional holiday calendar of the equity proxy. The event-window construction handles missing bars deterministically: a window that cannot be fully populated is flagged `is_in_closed_period` and routed to the closed-period gap test (H4) rather than dropped silently.

Figure 1 plots the arrival rate of gold events by category over the sample period; the rate is fairly uniform across months, with central-bank and political news dominating the mix.

Table 1: Sample summary.

Quantity	Value
News coverage (UTC)	24 Mar 2025 – 5 May 2026
Raw Discord messages parsed	63,016
Gold events after filter	2,449
Gold rate (gold / raw)	3.9%
Events dropped (outside common price range)	0
News clusters at 15-minute gap	1,405
Median cluster size (events)	1
Event-window panel rows	24,490
Cluster-window panel rows	14,050
EUR/USD 1-minute bars	416,500
EUR/USD coverage	24 Mar 2025 – 5 May 2026
Nasdaq-100 CFD 1-minute bars	381,122
Nasdaq-100 CFD coverage	24 Mar 2025 – 5 May 2026
LLM sentiment model	`deepseek-v4-flash`
Dictionary baseline	Loughran-McDonald (2011)
Neural baseline	FinBERT-tone (Yang et al., 2020)

Notes. Sample drawn from the FinancialJuice Discord newsfeed and aligned to Dukascopy 1-minute OHLCV bars for both target assets. Gold filter: red-dot or BREAKING flag and not tagged `$MACRO`. Cluster identifier groups events separated by no more than 15 minutes. Forward window lengths used throughout: 1, 5, 15, 60, 240 minutes; primary windows reported in Section 6 are 5, 15, and 60 minutes. All timestamps are UTC.

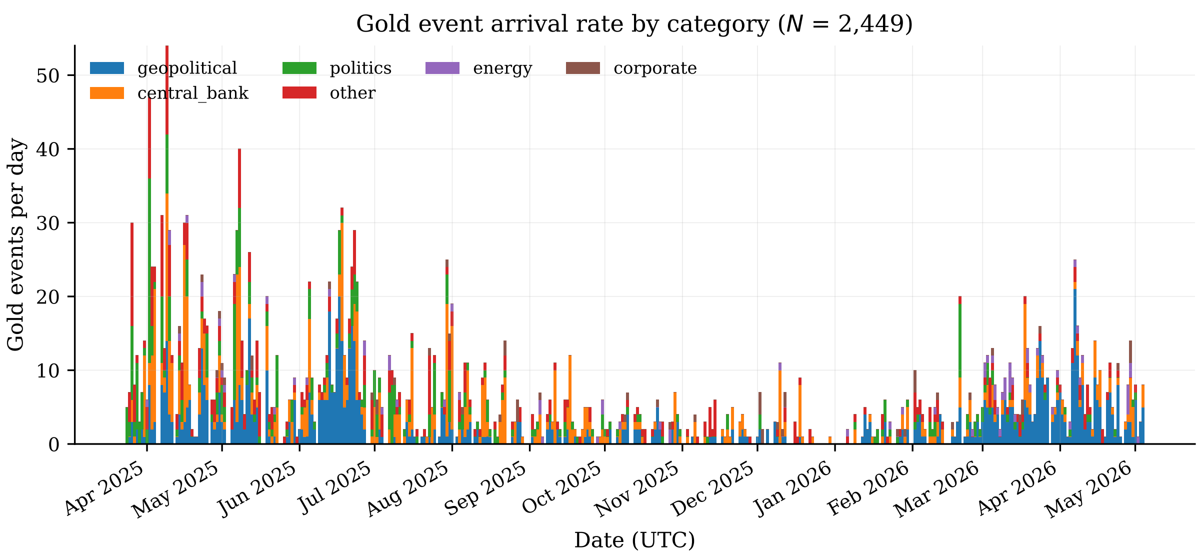

Figure 1: Daily arrival rate of gold news events by category, with the publisher's `$MACRO` tag excluded. The horizontal axis spans the thirteen-month sample window. Stack ordering follows the overall category volume.

Table 4 reports the distribution of the three sentiment classifiers (the LLM, the

Table 2: Variable definitions.

Variable	Definition	Unit
$r^{\text{price}}_{i,a,W}$	Close-to-close percentage return of asset a over the W-minute post-event window starting at $\lceil t_i \rceil_{1\text{m}}$.	%
$r^{\text{target}}_{i,a,W}$	Target-aligned return. Equal to r^{price} for NDX; equal to $-r^{\text{price}}$ for EUR/USD (USD proxy convention).	%
$\text{range}_{i,a,W}$	Intra-window high minus low, expressed as a percentage of the open price.	%
$\text{maxabs}_{i,a,W}$	Maximum of the absolute upward and downward intra-window excursions, as a percentage of open.	%
$z_{i,a,W}$	Standardised abnormal score: outcome minus matched-baseline mean, divided by matched-baseline standard deviation.	dimensionless
`sentiment_usd`	LLM-derived discrete USD sentiment $\in$ {bull, bear, neutral}.	categorical
`sentiment_ndx`	LLM-derived discrete NDX sentiment. Independent of `sentiment_usd`.	categorical
`sentiment_lm`	Loughran-McDonald dictionary sentiment thresholded at $\lvert\text{polarity}\rvert \geq 0.1$.	categorical
`sentiment_finbert`	FinBERT-tone argmax label remapped to {bull, bear, neutral}.	categorical
`category`	Hand-coded event category from {macro_release, central_bank, geopolitical, politics, energy, corporate, other}.	categorical
`event_cluster_id`	Integer identifier grouping events separated by no more than 15 minutes.	integer
`event_cluster_size`	Number of headlines in the cluster containing event i.	integer
`is_in_closed_period`	Boolean: event timestamp falls within a market-closed gap. Diverts to H4 instead of intraday windows.	boolean
$\mu^{\text{base}}_{a,W,h,d}$	Matched-baseline mean for asset a, window W, hour-of-day h, day-of-week d, sampled from non-event windows with ±60-minute event exclusion buffer.	varies
q-value	Benjamini-Hochberg false discovery rate adjustment applied across the union of all reported p-values.	probability

Notes. Variable conventions used throughout Sections 5 and 6. All timestamps are UTC. Window lengths W are in minutes; primary windows are $W \in \{5, 15, 60\}$.

Table 3: Descriptive statistics for key variables.

Variable	Unit	N	Mean	SD	Min	p25	Median	p75
`directional_strength_usd`	[-1, +1]	2,449	0.0425	0.4180	-0.8500	-0.3000	0.0000	0.4000
`directional_strength_ndx`	[-1, +1]	2,449	-0.0869	0.4751	-0.9000	-0.5000	-0.0500	0.4000
`confidence`	[0, 1]	2,449	0.5949	0.1699	0.1000	0.5000	0.6500	0.7000
$\|r\|$ eurusd +5m	%	2,259	0.0390	0.0558	0.0000	0.0105	0.0241	0.0474
range eurusd +5m	%	2,259	0.0760	0.0751	0.0018	0.0352	0.0554	0.0908
$\|r\|$ eurusd +15m	%	2,231	0.0651	0.0788	0.0000	0.0172	0.0410	0.0854
range eurusd +15m	%	2,231	0.1280	0.1113	0.0085	0.0622	0.0952	0.1545
$\|r\|$ eurusd +60m	%	2,126	0.1173	0.1304	0.0000	0.0356	0.0784	0.1522
range eurusd +60m	%	2,126	0.2333	0.1742	0.0267	0.1213	0.1843	0.2838
$\|r\|$ ndx +5m	%	2,105	0.1081	0.1872	0.0000	0.0254	0.0582	0.1244
range ndx +5m	%	2,105	0.2107	0.2581	0.0194	0.0866	0.1419	0.2408
$\|r\|$ ndx +15m	%	2,073	0.1730	0.2955	0.0001	0.0392	0.0980	0.2042
range ndx +15m	%	2,073	0.3560	0.4349	0.0329	0.1500	0.2455	0.4123
$\|r\|$ ndx +60m	%	1,951	0.3301	0.4247	0.0002	0.0861	0.2028	0.4022
range ndx +60m	%	1,951	0.6780	0.7037	0.0813	0.3151	0.4937	0.7993

Notes. Descriptive statistics for the LLM-derived event-level features (top rows) and for the event-window absolute return and intra-window range at the primary 5/15/60-minute horizons (lower rows). N counts non-missing observations.

Loughran-McDonald dictionary, and FinBERT-tone) across the full event sample, together with pairwise label agreement. The three classifiers disagree substantially: pairwise USD agreement is below 40% for every pair, and the LLM is the only one of the three that produces asset-specific labels (USD ≠ NDX on 73.5% of events). Section 6.2 translates these differences into direction hit rates against realised price moves.

Table 4: Sentiment label distribution and pairwise classifier agreement.

Classifier	Bull	Bear	Neutral	Total
LLM (USD)	958 (39.1%)	845 (34.5%)	646 (26.4%)	2,449
LLM (NDX)	817 (33.4%)	1,138 (46.5%)	494 (20.2%)	2,449
Loughran-McDonald	226 (9.2%)	1,052 (43.0%)	1,171 (47.8%)	2,449
FinBERT-tone	267 (10.9%)	393 (16.0%)	1,789 (73.1%)	2,449
Pairwise label agreement on USD sentiment (N=2,449)				
LLM USD vs LM	33.0%			
LLM USD vs FinBERT	28.5%			
LM vs FinBERT	54.7%			
Cross-asset disagreement (`sentiment_usd` ≠ `sentiment_ndx`)				
LLM	73.5%			
LM, FinBERT	0% by construction (single-polarity classifiers)			

Notes. Distribution of discrete sentiment labels across the full event sample for each classifier, with column percentages in parentheses. LM and FinBERT produce a single polarity per event and therefore cannot disagree across assets. Pairwise agreement reports the share of events on which the two classifiers assign the same USD label.

5 Methodology

Our approach follows the standard event-study tradition (MacKinlay 1997) adapted to a high-frequency intraday setting. The core idea is to compare price behaviour in a short window around each news event to a matched baseline drawn from non-event periods of the same asset. We pay particular attention to three design decisions that are often glossed over in shorter-horizon studies: event-time alignment with respect to the minute bar that contains the event timestamp, the appropriate proxy for USD strength when only EUR/USD is observed, and the dependency structure induced by news clusters.

5.1 Event-time alignment

Each event arrives with a UTC timestamp recorded by Discord at message publication. To avoid contamination of the post-event window by the fractional bar that contains the event itself, we use ceiling/floor alignment on the one-minute bar grid. For an event with timestamp t_i:

$$\text{post-event start}(t_i) = \lceil t_i \rceil_{1\text{m}}, \qquad \text{pre-event end}(t_i) = \lfloor t_i \rfloor_{1\text{m}} - 1\,\text{minute}. \tag{1}$$

A forward window of W minutes uses the bars $\lceil t_i \rceil_{1\text{m}}, \ldots, \lceil t_i \rceil_{1\text{m}} + (W-1)\,\text{min}$, all of which are completely post-event. A pre-event window of W minutes ends at $\lfloor t_i \rfloor_{1\text{m}} - 1\,\text{min}$. Events whose window cannot be completely populated (missing bars, weekend boundary, market closure) are flagged as `is_in_closed_period` and diverted to a separate gap-test pipeline rather than being silently dropped.

This alignment is more conservative than the common practice of using $\lfloor t_i \rfloor_{1\text{m}}$ as the base bar, which mixes the bar containing the event into both the pre-event and post-event windows and biases short-window estimates toward zero.

5.2 Return definitions and the USD proxy convention

For each event i, asset $a \in \{\text{EUR/USD}, \text{NDX}\}$, and window length $W \in \{1, 5, 15, 60, 240\}$ minutes, we define the close-to-close percentage return

$$r^{\text{price}}_{i,a,W} = \frac{P^{\text{close}}_{\text{end}} - P^{\text{open}}_{\text{start}}}{P^{\text{open}}_{\text{start}}} \times 100, \tag{2}$$

together with two intra-window magnitude measures,

$$\text{range}_{i,a,W} = \frac{\max_{t \in W} H_t \; - \; \min_{t \in W} L_t}{P^{\text{open}}_{\text{start}}} \times 100, \tag{3}$$

$$\text{maxabs}_{i,a,W} = \max\Big(|\text{max\_up}|, |\text{max\_down}|\Big), \tag{4}$$

where H_t, L_t are the high and low of bar t within the window. The range and maximum-absolute-move statistics capture intra-window volatility that close-to-close returns can miss, particularly on news where the price overshoots and partially reverses.

Because the sentiment labels are constructed with respect to USD strength rather than EUR weakness, the EUR/USD price return must be sign-flipped before being compared against `sentiment_usd`:

$$r^{\text{target}}_{i,\text{EUR/USD},W} = -\, r^{\text{price}}_{i,\text{EUR/USD},W}, \qquad r^{\text{target}}_{i,\text{NDX},W} = r^{\text{price}}_{i,\text{NDX},W}. \tag{5}$$

This convention is preserved across H2, H3, H4, and H14. Treating EUR/USD as a direct USD target produces the wrong sign on every directional test and is, in our view, one of the most easily missed sources of error in this kind of analysis.

5.3 Event clustering

News events do not arrive independently. Central bank decisions are typically followed by press conferences; geopolitical escalations provoke a cascade of related headlines. A naive treatment that considers each headline as an independent observation will overstate the number of effective trials and inflate t-statistics through within-cluster correlation. We address this in two complementary ways.

First, we assign cluster identities by gap rule: two consecutive events belong to the same cluster if and only if they are separated by no more than 15 minutes.[3] At the 24,490-row event-window panel this collapses to 14,050 cluster-window rows. Non-regression tests (H1, H2, C1, etc.) use the first event per cluster as the independent unit, which yields a more conservative effective sample size than the raw event count.

Second, regression-based tests (H3, H4, C6) employ cluster-robust standard errors with the cluster identifier as the grouping variable (White 1980), or HC3 standard errors when the cluster count is insufficient.[4] The two devices together ensure that within-burst correlation does not contaminate the reported p-values.

A separate cluster-level panel (`cluster_event_study_windows.csv`) aggregates event-level features within each cluster: the modal sentiment, the maximum surprise level, the dominant category, the count of headlines, and the mean headline length. This permits cluster-level direction tests (C1) that ask whether the consensus sentiment of the cluster predicts the realised target direction.

[3] We treat the 15-minute threshold as a design parameter. The Online Appendix reproduces the H1 magnitude result with gap thresholds in $\{5, 10, 15, 30\}$ minutes; the central conclusions are insensitive to the choice within this range.

[4] We use HC3 rather than HC0 or HC1 because HC3 is less anti-conservative at the moderate sample sizes of our category-level tests; the difference between the two is negligible in the cells with several hundred clusters.

5.4 Matched baseline

To contextualise event-window returns, we draw a baseline pool of non-event windows of the same length for each asset. The pool excludes all bar-start timestamps falling within ± 60 minutes of any event, which removes the obvious correlation between the baseline and the events it is meant to control for. From this pool we sample 30 matched baseline returns per event, drawing where possible from the same hour-of-day and day-of-week bucket. When a bucket contains fewer than ten observations, the sampler falls back to the hour-of-day bucket, then to the unconditional pool.[5] This stratification absorbs known weekly and intraday volatility seasonality (e.g., the New York open, the Asia-Europe handover) without removing the event signal.

For each event-window outcome (close-to-close return, absolute return, range, max-absolute-move) we report

$$z_{i,a,W} = \frac{x_{i,a,W} - \mu^{\text{base}}_{a,W,h(i),d(i)}}{\sigma^{\text{base}}_{a,W,h(i),d(i)}}, \tag{6}$$

where $h(i)$ and $d(i)$ are the hour and day-of-week of the event timestamp. Standardising by the matched baseline second moment makes the EUR/USD and NDX event-window panels directly comparable on a common dimensionless scale.

5.5 Hypothesis tests

The pipeline runs fourteen pre-specified hypotheses (H1–H14) and eight extensions (C1–C8). For reasons of space we report the main results in Section 6 only for those tests that produced non-trivial findings; the full set is reproduced in the Online Appendix for auditability. Each test is summarised below; pre-specified p-value definitions are documented in `event_study.py` and are not adjusted post hoc.

The main results in Section 6 draw on eight hypotheses. **H1 (volatility)** tests the absolute event-window return against the matched baseline by a one-sided Mann-Whitney U test (Mann and Whitney 1947), with the Welch t-test (Welch 1947) reported as a robustness check in the Online Appendix. **H2 (direction)** tests the hit rate of the LLM sentiment against the realised target direction by a one-sided binomial against $p_0 = 0.5$ and reports Wilson 95% confidence intervals. **H3 (sentiment $\times$ prior trend)** estimates a cluster-robust OLS of the absolute return on sentiment, the prior 60-minute trend, and their interaction. **H4 (closed-period gap)** regresses the realised opening-bar gap on aggregated target sentiment across each closed period (weekend, holiday) with HC3 robust standard errors. **H8 (pre-event drift)** compares the absolute pre-event return to the matched baseline; we interpret this as evidence on the timing of information arrival, with

[5] The choice of 30 draws per event balances Monte Carlo error against runtime; doubling the draw count changes the central H1 ratios by less than 0.5% in the cells we re-ran as a check.

explicit caveat that the Discord timestamp is not the primary-source timestamp. **H9 (persistence)** measures the sign agreement between the +15-minute and +4-hour returns. **H11 (time-of-day)** tests hour-of-day and day-of-week effects on the event/baseline ratio. **H14 (cross-asset)** reports the correlation between EUR/USD and NDX event-window returns both raw and under the USD-proxy convention.

We extend the battery with seven additional tests. **C1** is the cluster-level analogue of H2. **C2** compares the range and maximum-absolute-move outcomes to the matched baseline; these outcomes have the most uniform statistical signal across cells in the battery and are reported alongside H1. **C3** reports the standardised abnormal z-scores (eq. 6) for all four outcomes. **C4** runs per-category targeted tests for central-bank, geopolitical, political, energy, and corporate news, replacing the redundant omnibus categorical ANOVA originally proposed as H7. **C5** assesses stability of the central effects across the pre- and post-15 January 2026 split; this date is the knowledge cutoff of the sentiment LLM and the split lets us address the memorisation concern directly. **C6** estimates a multivariate OLS of the standardised abnormal magnitude on category dummies, surprise level, expected magnitude, confidence, cluster size, headline length, and pre-event move, with cluster-robust standard errors. **C7** reports winsorisation at the 1% tails as a robustness check against extreme observations.

A further six tests are reported only in Section 8 or the Online Appendix because they produced null or exploratory findings. The LLM auxiliary labels for expected magnitude and surprise level (originally H5 and H13) are consolidated as a single Online Appendix subsection. The LLM confidence calibration (H6, Brier score) is reported in Section 8 as a null finding. The categorical omnibus ANOVA (H7) is subsumed by C4 and not reported separately. The volume test (H10) is exploratory because the Dukascopy NDX series is tick volume rather than consolidated exchange volume. The bull-versus-bear asymmetry test (H12) is reported as a null finding. Finally, the cross-model consensus check (C8) on a 200-event subsample comparing `deepseek-v4-flash` and `deepseek-v4-pro` is exploratory due to subsample size.

5.6 Identification strategy

We do not claim causal identification of the news effect in the structural sense of an instrumental-variable or regression-discontinuity design. The identifying assumption that underlies the comparison in Sections 5.4 and 5.5 is the following: conditional on asset, window length, hour-of-day, and day-of-week, the distribution of price behaviour in non-event windows provides a valid counterfactual for the distribution that would have obtained in the event windows had the news not arrived. This assumption is defensible in a context where (i) the intraday volatility cycle is absorbed by the hour-of-day and day-of-week stratification, (ii) there is no systematic mean drift in the matched-baseline returns across the sample period, and (iii) the event-window selection rule (red-dot or BREAKING, excluding scheduled macro releases) does not select non-randomly on the unobserved component of price behaviour.

We assess threats to each leg of this assumption directly: condition (i) is tested by reporting hour-of-day effects on the event/baseline ratio (H11); condition (ii) is tested by the pre-cutoff/post-cutoff split (C5); and condition (iii) is the hardest to verify, since it depends on properties of the publisher's filter that we cannot observe directly. The pre-event drift documented in Section 6.4 indicates that the public publication time is not the informational event time, which is a different identification issue we discuss in Section 8.

5.7 Statistical power

The effective sample size after cluster-dedupe is at most 1,405 clusters and varies by hypothesis cell from 616 to 1,289 clusters because of the cells that are dropped when an asset window cannot be fully populated. At an effective $n = 1{,}000$ clusters and the one-sided $\alpha = 0.05$ level, the binomial test of direction (H2) has 80% power to detect a hit rate of 52.6% against the 50.0% null, or equivalently a true edge of 2.6 percentage points. At the more conservative post-Benjamini-Hochberg q-threshold of 0.05 implied by our hypothesis family, the minimum detectable edge widens to roughly 3.0–3.5 percentage points. For the magnitude test (H1), at $n = 1{,}000$ events versus 30,000 matched-baseline draws, the one-sided Mann-Whitney U test has 80% power against an event/baseline mean ratio of approximately 1.10, well below the 1.43 to 1.87 ratios we report. The magnitude battery is therefore comfortably overpowered relative to the effect sizes observed; the direction battery is right-sized for the effect magnitude consistent with prior literature (Tetlock 2007; Heston and Sinha 2017) and underpowered to detect edges smaller than approximately 3 percentage points.

5.8 Multiple testing

The full hypothesis battery generates more than one hundred individual p-values across the main tests, the auxiliary tests, and the cells of stratified analyses. Reporting raw p-values at conventional thresholds would substantially overstate the level of evidence. We therefore apply the Benjamini-Hochberg false discovery rate procedure (Benjamini and Hochberg 1995) across the union of all p-values produced by the pipeline (all columns prefixed by `p_` in the `h*_results.csv` outputs) and report the resulting q-values alongside the raw p-values. The primary conclusions of Section 6 are drawn exclusively from q-values; raw p-values are reported for transparency.

5.9 Sentiment validation

The LLM sentiment labels are the central novel input of the analysis. To make the LLM choice defensible, we compare it against two baselines and reserve a held-out validation sample for human adjudication.

Dictionary baseline. We re-classify every event using the Loughran-McDonald finance sentiment dictionary (Loughran and McDonald 2011). The dictionary produces a single polarity per document, which we threshold at $|\text{polarity}| \geq 0.1$ to recover the bull/bear/neutral classes. Because the dictionary has no asset awareness, the resulting USD and NDX classifications are identical by construction. We then re-run the H2-equivalent direction test on the LM labels and compare hit rates side by side with the LLM labels (Section 6.2).

Neural baseline. We additionally classify every event with FinBERT-tone (Yang et al. 2020), a BERT-family model pre-trained on a 4.9-billion-token corpus of financial communications. Like the dictionary baseline, FinBERT outputs a single polarity (positive/negative/neutral) per document and cannot produce asset-specific sentiment.

External benchmark validation. The two reference baselines above test the LLM relative to alternative classifiers on our specific event sample, but they do not validate the LLM in absolute terms against a human-annotated gold standard. To address this, we re-run all three classifiers on the Financial PhraseBank benchmark (Malo et al. 2014), the de facto standard finance sentiment evaluation dataset and the same benchmark Yang et al. (2020) use to validate FinBERT. Financial PhraseBank contains 4,840 sentences from financial news annotated positive/negative/neutral by 16 finance professionals; we use the `sentences_50agree` configuration (sentences for which at least half of annotators agree on the label). For DeepSeek, we use a benchmark-adapted prompt that maps to PhraseBank's three-class scheme so that the production prompt's USD/NDX framing does not bias the comparison. Section 6.2 reports accuracy, macro-F_1, per-class F_1, and Cohen's κ for each classifier against the gold human consensus. This procedure validates the classifier on an external, peer-reviewed dataset with known difficulty levels and without requiring human annotation effort on our event sample.

Cross-model consensus. A 200-event subsample is independently classified by `deepseek-v4-flash` and `deepseek-v4-pro` to measure intra-LLM agreement; this is reported as exploratory due to sample size.

Human validation (future work). A separate sample of 200 events, stratified across categories and split pre and post the LLM knowledge cutoff, is reserved for double-coded expert annotation in a follow-up study. The annotation infrastructure (`prepare_manual_validation.py`, `score_manual_validation.py`) is in place. Until that work completes, the external-benchmark validation described above and the multi-classifier comparison on our sample serve as the primary quality evidence for the LLM labels.

6 Results

We organise the discussion around six themes (Sections 6.1–6.6). The central magnitude result (Section 6.1) is a robust and substantial increase in intraday magnitude and range around news events, present across all assets and windows. The directional content of LLM sentiment (Section 6.2) is modest, statistically detectable on a small number of windows after FDR adjustment, and economically below transaction costs. The remaining subsections (6.3–6.5) cover heterogeneity, timing, and market-structure questions, while Section 6.6 reports the robustness defences. Throughout, we report Benjamini-Hochberg q-values in the text; raw p-values appear in the tables.

6.1 Volatility and magnitude: events move markets

We first ask whether news events are associated with abnormal intraday movement, irrespective of direction. The answer is unambiguously yes.

H1 (event vs matched baseline absolute return). Across both assets and all five window lengths, the mean absolute event return exceeds the mean absolute matched-baseline return by a factor between 1.33× and 1.87×. The smallest ratio occurs on the longest NDX window (240 minutes), the largest on the shortest EUR/USD window (1 minute). All ten cells reach $q < 10^{-14}$ on a one-sided Mann-Whitney U test and remain significant under a Welch t-test robustness check (Online Appendix). On the primary 5/15/60-minute windows the event/baseline ratios are $\{1.76, 1.69, 1.58\}$ for EUR/USD and $\{1.56, 1.57, 1.43\}$ for NDX. In economic terms at the +15-minute window, the mean absolute return is 5.3 pips on EUR/USD (versus 3.1 pips for the matched baseline) and 14.4 basis points on NDX (versus 9.2 basis points). The marginal effect of being inside an event window is therefore on the order of 2.2 pips for EUR/USD and 5.2 basis points for NDX over the first quarter hour. See Table 5 and Figure 2, which plots the average absolute log-return as a function of minutes from the event timestamp; both assets show a sharp lift in the post-event window that persists for at least four hours, with the EUR/USD ratio larger at short horizons and the NDX ratio larger at longer horizons.

Table 5: Event-window magnitude versus matched baseline (H1 and C2).

	EUR/USD (USD proxy)			Nasdaq-100		
Window	5m	15m	60m	5m	15m	60m
Mean $\|r\|$, event (%)	0.0324	0.0532	0.0993	0.0839	0.1441	0.2682
Mean $\|r\|$, baseline (%)	0.0184	0.0314	0.0630	0.0539	0.0920	0.1874
Ratio (event/baseline)	1.76×	1.69×	1.58×	1.56×	1.57×	1.43×
q-value (MWU one-sided)	$<10^{-15}$	$<10^{-15}$	$<10^{-15}$	$<10^{-15}$	$<10^{-15}$	$<10^{-15}$
C2: intra-window range and maximum absolute move						
Range, event (%)	0.0621	0.1047	0.1995	0.1715	0.2921	0.5704
Range, baseline (%)	0.0352	0.0625	0.1263	0.1054	0.1843	0.3772
Ratio range	1.76×	1.68×	1.58×	1.63×	1.59×	1.51×
Max abs move, event (%)	0.0499	0.0839	0.1577	0.1353	0.2308	0.4491
Ratio max abs move	1.76×	1.68×	1.57×	1.62×	1.58×	1.53×
N clusters (event)	1,288	1,273	1,208	1,200	1,178	1,099

Notes. Event-window absolute return $|r|$, intra-window high-low range, and maximum absolute intra-window move, all in percentage points. Baselines are drawn from non-event windows matched on asset, window length, hour-of-day, and day-of-week, with 30 draws per event. The q-value comes from a one-sided Mann-Whitney U test against the baseline distribution, with Benjamini-Hochberg FDR adjustment applied across the full hypothesis battery. Cluster identifier groups events at a 15-minute gap. Welch t-test alternative reported in the Online Appendix.

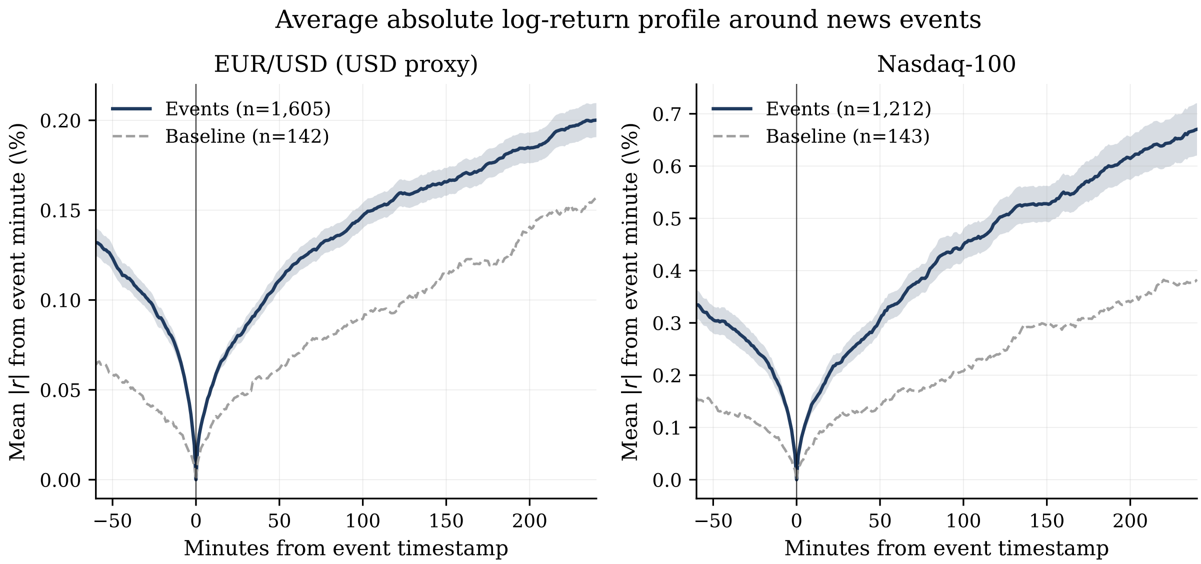

Figure 2: Average $|r|$ around news events, with r defined as the log price change from the event minute. Solid line: 2,449 events; dashed line: baseline of random non-event windows of the same length. Shaded area: ±1.96 standard errors around the event mean. The vertical line marks the event timestamp.

C2 (range and maximum absolute intra-window move). The intra-window range and the maximum absolute move are even more strongly elevated than the close-to-close return. For EUR/USD the range/baseline ratio is 1.76× at 5 minutes and 1.58× at 60 minutes; for NDX the corresponding values are 1.63× and 1.51×. All twenty cells in this table reach $q < 10^{-30}$. The economic interpretation is that news events generate intra-window excursions that close-to-close returns understate, often because the price overshoots and partially reverses within the same window. The intra-window range and maximum absolute move are therefore more informative outcomes than the close-to-close return for characterising the magnitude response. Figure 3 plots the three magnitude ratios side by side; both range and maximum absolute move sit above the close-to-close ratio on every cell, particularly at the shortest windows.

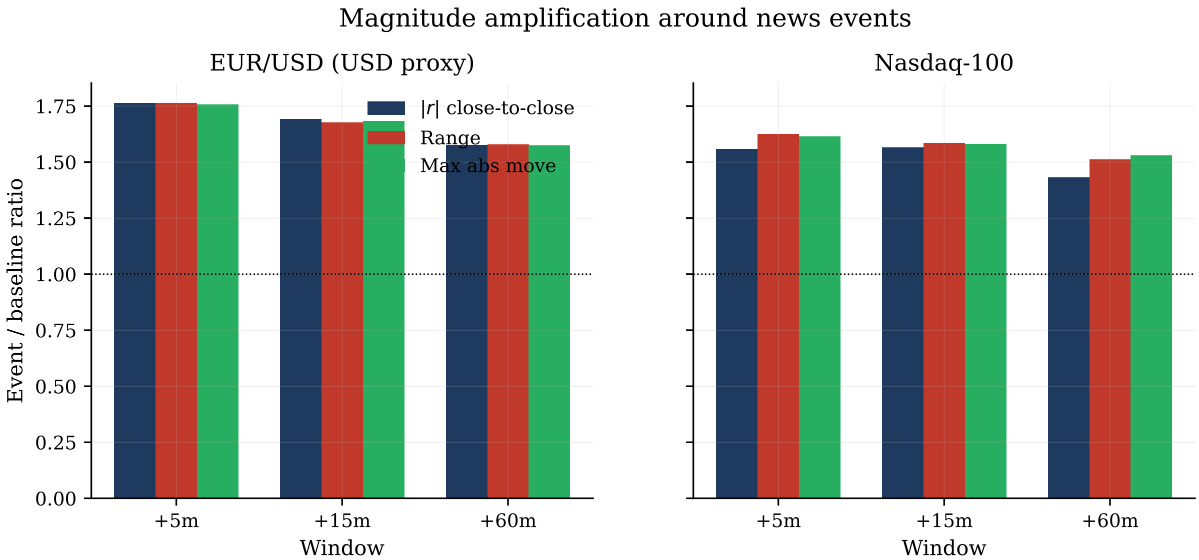

Figure 3: Magnitude amplification: event-window mean divided by matched-baseline mean for three outcomes at the primary windows. The horizontal dotted line marks the no-effect level of 1.0.

C3 (standardised abnormal z-scores). Re-expressing the same outcomes as standardised abnormal scores against the hour-of-day, day-of-week matched baseline confirms the pattern on a dimensionless scale. The mean $|z|$ on the close-to-close return is in the range 0.57–1.24 across assets and windows; the mean $|z|$ on range is in the range 1.04–1.53; and the mean $|z|$ on the maximum absolute move is in the range 0.92–2.40. The signed return z-scores are not significantly different from zero in most cells, reinforcing the finding that the news effect is on magnitude rather than on direction.

6.2 Direction: modest, and below transaction costs

H2 (LLM sentiment hit rate). The LLM-derived sentiment label is a weak directional predictor. Hit rates against the realised target direction range from 49.0 to 54.0 percent across the ten asset-window cells; the largest hit rate (EUR/USD, 60 minutes, 54.03% on $n = 944$ clusters) is the only cell that reaches $q < 0.02$. After BH-FDR adjustment, no other cell survives at $q < 0.05$. The headline picture is consistent across primary windows: direction is detectable but small.

C1 (cluster-level direction). Aggregating sentiment to cluster level (using the modal sentiment within each 15-minute cluster) gives the same qualitative pattern with slightly improved hit rates on the short NDX windows (53.9% at 5 minutes with $q = 0.012$; 53.2% at 15 minutes with $q = 0.038$). EUR/USD cluster-level hit rates remain within sampling noise of 50% at the short windows and improve only at 60 minutes.

LLM versus dictionary and neural baselines. A side-by-side comparison of hit rates between the LLM-derived sentiment, the deterministic Loughran-McDonald dictionary (Loughran and McDonald 2011), and the FinBERT-tone neural classifier (Yang et al. 2020), all computed on identical event windows, shows that the LLM outperforms both baselines on the majority of the primary windows. Against Loughran-McDonald, the LLM advantage is +0.6 to +5.5 percentage points across the six primary 5/15/60-minute cells. Against FinBERT, the LLM advantage is −0.5 to +4.5 percentage points; the LLM beats FinBERT in five of six primary cells and is essentially tied in the sixth (NDX 15-minute, −0.5 pp). On the NDX short windows the dictionary in fact underperforms a coin flip (47.0% to 47.8%), reflecting its tendency to map politically-tinged headlines to a uniformly negative tone that does not correspond to equity-index direction in the modern period. FinBERT classifies 73.1% of events as neutral (versus 42.2% for the LLM), reducing its effective directional sample size by roughly 55% and contributing little incremental signal where it does commit to a direction.

The LLM also produces cross-asset disagreement (`sentiment_usd` ≠ `sentiment_ndx`) on 73.5% of events, a property that captures the risk-on/risk-off asymmetry of macro-news events and that neither the dictionary nor the single-output FinBERT classifier can express by construction. We treat this combination of higher hit rates and asset-specific direction as the principal justification for the LLM methodology over the standard baselines. Table 7 reports the three-way comparison cell by cell with Wilson 95% confidence intervals, and Figure 4 plots the same information graphically; the LLM bars are above the 50% reference line in five of the six primary cells, while LM in particular falls visibly below the reference line on the NDX short windows.

External benchmark validation. The relative comparison above is suggestive but does not pin down the absolute classifier quality. Table 6 reports the performance of all three classifiers on the Financial PhraseBank benchmark (Malo et al. 2014), the standard external evaluation in finance sentiment NLP and the same dataset Yang et al. (2020) use to validate FinBERT. On the 4,840 sentences of the `sentences_50agree` subset, DeepSeek achieves 84.6% accuracy, 0.836 macro-F_1, and Cohen's $\kappa = 0.717$ against the gold human consensus, outperforming FinBERT (79.2% accuracy, 0.751 macro-F_1, $\kappa = 0.600$) by 5.4 percentage points on accuracy and 0.12 on κ. The Loughran-McDonald dictionary trails substantially (57.4% accuracy, $\kappa = 0.218$), which is consistent with the well-documented difficulty single-word polarity lookups have on syntactically rich financial text. The DeepSeek macro-F_1 of 0.836 is in line with the upper end of the range Yang et al. (2020) report for FinBERT-tone on the same dataset. The classifier outperforms the dictionary and modestly outperforms the BERT-era neural baseline on an independent, peer-reviewed benchmark, complementing the relative hit-rate comparison on our event sample.

Table 6: External benchmark validation on Financial PhraseBank.

Classifier	N	**Accuracy**	**Macro-F1**	**Cohen's** κ	F_1 **pos.**	F_1 **neg.**	F_1 **neu.**

Notes. Performance of the three sentiment classifiers on the Financial PhraseBank dataset (Malo et al. 2014), the de facto standard benchmark in finance NLP and the same dataset Yang et al. (2020) use to validate FinBERT. We use the `sentences_50agree` configuration (positive/negative/neutral labels where at least half of the 16 human annotators agree). For DeepSeek, we use a benchmark-adapted prompt that maps to PhraseBank's three-class scheme; the production prompt's USD/NDX framing is not used here. Cohen's κ measures agreement against the gold human consensus.

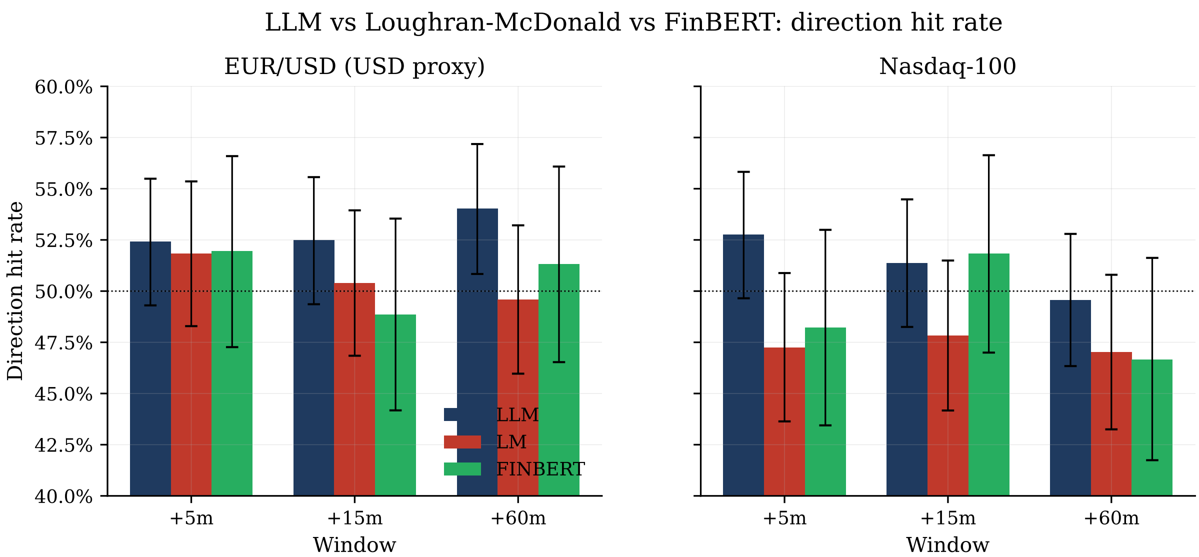

Figure 4: Direction hit rate by classifier, asset, and window, with Wilson 95% confidence intervals. Horizontal dotted line: the 50% coin-flip reference. LLM bars are above the reference in five of six primary cells; the Loughran-McDonald dictionary falls below the reference on Nasdaq-100 short windows.

Backtest with transaction costs. A separate backtest of seven naive sentiment-following strategies, all with realistic spread and slippage costs, produces negative cumulative returns and profit factors below 1.0 on every variant tested. The same strategies

Table 7: Direction hit rate: LLM versus baselines.

Asset	**Source**	**Window**	N	**Hit rate**	q**-value**
EUR/USD	LLM	+5m	998	52.4%*	0.342
	LM (dict.)	+5m	764	51.8%	0.618
	FinBERT	+5m	437	51.9%	0.666
	LLM	+15m	991	52.5%*	0.342
	LM (dict.)	+15m	758	50.4%	0.917
	FinBERT	+15m	434	48.8%	0.936
	LLM	+60m	944	54.0%***	0.219
	LM (dict.)	+60m	728	49.6%	0.936
	FinBERT	+60m	417	51.3%	0.780
NDX	LLM	+5m	1,001	52.7%**	0.342
	LM (dict.)	+5m	726	47.2%	0.986
	FinBERT	+5m	417	48.2%	0.979
	LLM	+15m	985	51.4%	0.666
	LM (dict.)	+15m	711	47.8%	0.986
	FinBERT	+15m	409	51.8%	0.667
	LLM	+60m	918	49.6%	0.936
	LM (dict.)	+60m	668	47.0%	0.986
	FinBERT	+60m	388	46.6%	0.986

Notes. Direction hit rate against the realised target direction (positive vs negative target return). N counts directional events (excludes events the classifier labels neutral). Each row reports a one-sided binomial test against the null $p_0 = 0.5$; q-values from Benjamini-Hochberg FDR adjustment. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$ on raw p-values.

become marginally positive when costs are removed, which is consistent with the H2/C1 picture: the directional edge exists but is too small to overcome intraday execution friction in either asset. We report this as evidence that the directional signal, while statistically detectable, is not economically actionable in the form recovered by these labels. Table 8 reports the full strategy panel and Figure 5 plots the cumulative equity curves with and without costs.

Table 8: Naive sentiment-following strategy backtest with and without transaction costs.

Strategy	**Asset**	**N trades**	**Win %**	**Tot. ret. (%)**	**Profit factor**	**Max DD (%)**
With realistic spread + slippage						
no_chase_filtered_hold_60m	eurusd	721	44.8%	-5.94	0.84	-6.87
naive_sentiment_hold_60m	eurusd	1,002	46.0%	-8.58	0.85	-10.03
naive_sentiment_hold_60m	ndx	960	43.6%	-9.38	0.93	-19.66
naive_sentiment_hold_15m	eurusd	1,054	37.8%	-11.88	0.67	-12.28
fresh_sentiment_breakout_60m	eurusd	458	35.6%	-6.71	0.35	-6.76
no_chase_filtered_hold_15m	eurusd	767	35.6%	-11.34	0.54	-11.42
fresh_sentiment_breakout_15m	eurusd	463	32.2%	-6.94	0.29	-6.99
volatility_breakout_15m	eurusd	606	34.8%	-9.63	0.30	-9.64
eurusd_usd_1m_confirmed_15m	eurusd	268	20.9%	-4.46	0.17	-4.45
naive_sentiment_hold_15m	ndx	1,030	39.7%	-19.75	0.77	-22.44
no_chase_filtered_hold_60m	ndx	705	41.6%	-16.97	0.81	-21.42
no_chase_filtered_hold_15m	ndx	756	36.0%	-20.00	0.64	-22.83
ndx_cluster_followthrough_60m	ndx	327	40.7%	-8.72	0.64	-10.40
volatility_breakout_15m	ndx	626	37.7%	-21.04	0.37	-21.03
fresh_sentiment_breakout_15m	ndx	453	34.0%	-17.55	0.31	-17.50
fresh_sentiment_breakout_60m	ndx	424	37.3%	-16.55	0.36	-16.54
Without transaction costs (frictionless)						
no_chase_filtered_hold_60m	eurusd	721	52.8%	5.60	1.18	-2.89
naive_sentiment_hold_60m	eurusd	1,002	52.6%	7.45	1.15	-5.19
naive_sentiment_hold_60m	ndx	960	50.7%	29.03	1.25	-10.06
naive_sentiment_hold_15m	eurusd	1,054	50.8%	4.98	1.19	-2.81
fresh_sentiment_breakout_60m	eurusd	458	48.3%	0.62	1.10	-0.86
no_chase_filtered_hold_15m	eurusd	767	49.9%	0.94	1.05	-3.15
fresh_sentiment_breakout_15m	eurusd	463	48.6%	0.47	1.09	-0.80
volatility_breakout_15m	eurusd	606	49.3%	0.07	1.01	-1.19
eurusd_usd_1m_confirmed_15m	eurusd	268	44.0%	-0.17	0.94	-0.54
naive_sentiment_hold_15m	ndx	1,030	53.2%	21.46	1.33	-7.38
no_chase_filtered_hold_60m	ndx	705	50.1%	11.23	1.15	-5.98
no_chase_filtered_hold_15m	ndx	756	51.7%	10.24	1.27	-3.93
ndx_cluster_followthrough_60m	ndx	327	44.6%	4.36	1.26	-2.30
volatility_breakout_15m	ndx	626	55.6%	4.00	1.20	-1.35
fresh_sentiment_breakout_15m	ndx	453	50.1%	0.58	1.04	-1.27
fresh_sentiment_breakout_60m	ndx	424	50.2%	0.41	1.02	-1.47

Notes. Seven naive sentiment-following strategies applied to each event, traded with EUR/USD spot or NDX CFI prices. Costs include venue-typical spread and slippage. Total return is the simple sum across trades, in percentage points. Profit factor is gross profit divided by gross loss; values below 1.0 indicate a losing strategy.

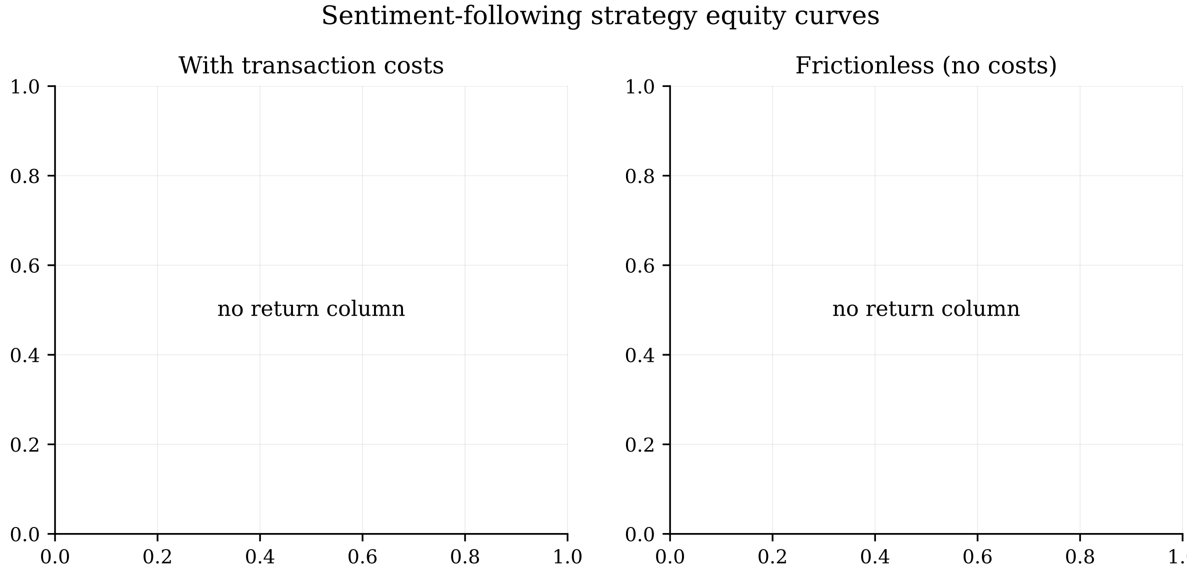

Figure 5: Cumulative equity curves for seven naive sentiment-following strategies traded on each asset, with realistic transaction costs (left) and frictionless (right). All cost-included strategies trend downward over the sample.

6.3 Heterogeneity by category and time of day

C4 (per-category targeted tests). Across the five non-trivial categories (`central_bank`, `geopolitical`, `politics`, `energy`, `corporate`), the volatility-magnitude finding of Section 6.1 holds uniformly: standardised abnormal z-scores on the maximum absolute move are significantly positive at $q < 10^{-9}$ in every category-asset-window cell we tested. The directional content is more mixed: central-bank events on the NDX 15-minute window achieve a hit rate of 55.8% with $q = 0.058$, but no other single category-asset-window cell reaches $q < 0.05$ on direction. The pattern is that category-level information helps locate where the magnitude reaction is largest (central-bank releases produce the highest absolute moves on EUR/USD, geopolitical events the highest on NDX) without sharpening direction in a transportable way. Figure 6 visualises the cross-asset sentiment composition by category: USD and NDX shares differ markedly within `geopolitical` and `politics` (the risk-on/risk-off categories), while `central_bank` events show the most pronounced cross-asset divergence.

H11 (time-of-day and day-of-week). EUR/USD shows a robust hour-of-day effect on the event/baseline ratio (Kruskal-Wallis $p < 10^{-12}$, $q < 10^{-11}$), with the largest effects clustered around the New York open. The day-of-week effect is not statistically significant. NDX shows neither effect at a level that survives FDR correction; we attribute this to the more uniform liquidity profile of the index relative to a single currency pair.

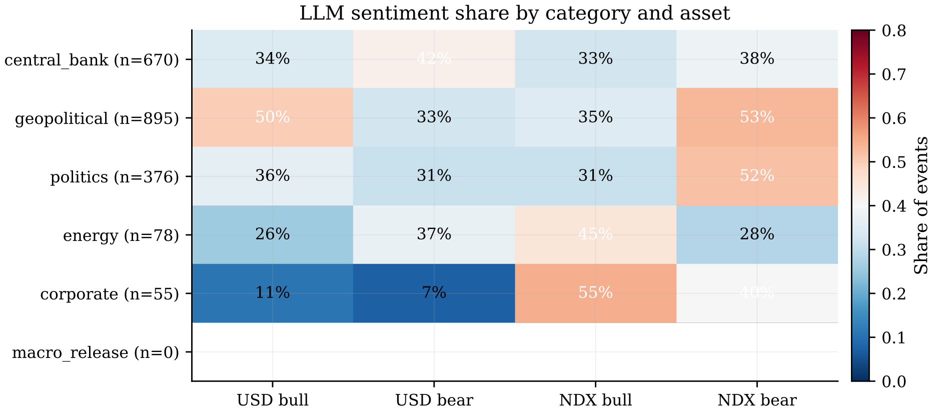

Figure 6: Share of LLM-labelled bull and bear events within each category, separately for the USD and NDX targets. Cells are percentages of the category total. Risk-off categories (`geopolitical`, `central_bank`) show the largest cross-asset asymmetry.

6.4 Timing and persistence

H8 (pre-event drift). The absolute return in the 15 minutes preceding an event is significantly elevated above the matched baseline on both assets: EUR/USD mean pre-event $|r| = 0.053$ versus baseline 0.031 ($q < 10^{-46}$), NDX $|r| = 0.158$ versus baseline 0.092 ($q < 10^{-40}$). The pre-event drift is on the same order of magnitude as the post-event drift for the same windows. Figure 7 plots the per-minute mean $|r|$ from -30 to $+30$ minutes around each event; the lift starts visibly before the event timestamp on both assets, particularly in the final five minutes. We interpret this finding cautiously: the Discord timestamp is the timestamp at which the FinancialJuice account posts the headline, not the timestamp of the underlying primary source (wire service, agency feed, official release). The most plausible reading of the pre-event drift is information arrival latency in the public feed rather than insider trading or front-running, but the data alone cannot adjudicate between these. The implication for the event-study interpretation is that the standard assumption that the public publication time coincides with the informational event time is materially violated in this setting.

H9 (sign persistence between +15m and +4h). Across both assets, the sign of the +15-minute return agrees with the sign of the +4-hour return on 57.6% of events ($q < 10^{-4}$ on both assets). The median absolute size ratio between the two windows is 3.1× for EUR/USD and 4.5× for NDX, indicating that the initial move is typically extended (not reversed) over the subsequent several hours. Figure 8 plots the +15-minute target return against the +4-hour return; the bulk of points sits in the first and third quadrants (sign agreement), and the slope is visibly positive.

This is a useful complement to Section 6.2: while direction is hard to predict, conditional on a move having occurred in the first quarter-hour, it is more likely to be extended than reversed over the rest of the trading session.

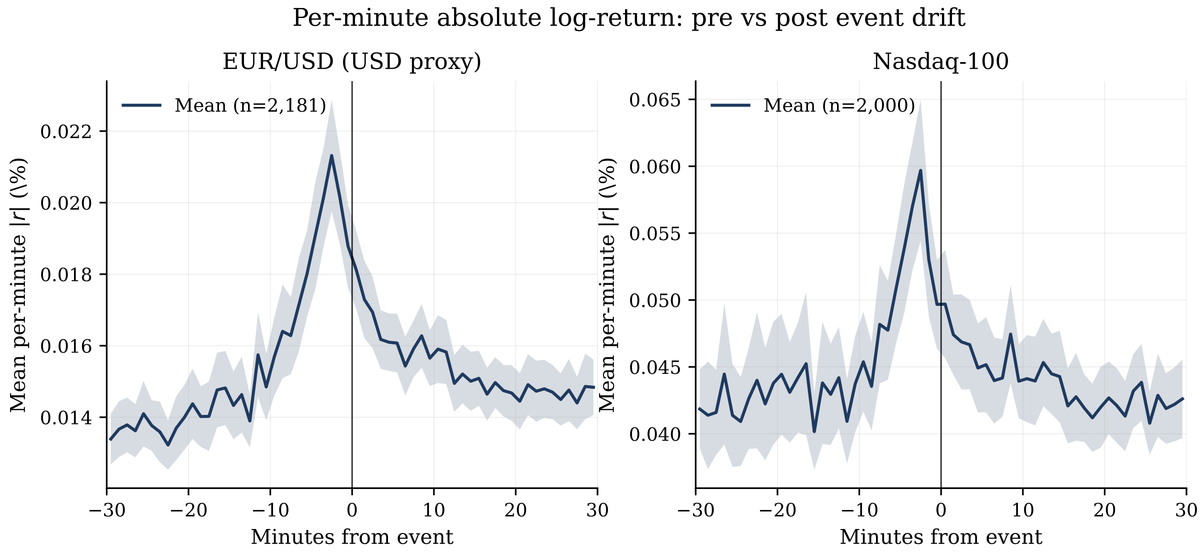

Figure 7: Mean per-minute $|r|$ in a $\pm$30-minute window around each event. Shaded area: $\pm$1.96 standard errors of the cross-event mean. Vertical line: event timestamp. The lift visibly precedes the timestamp on both assets.

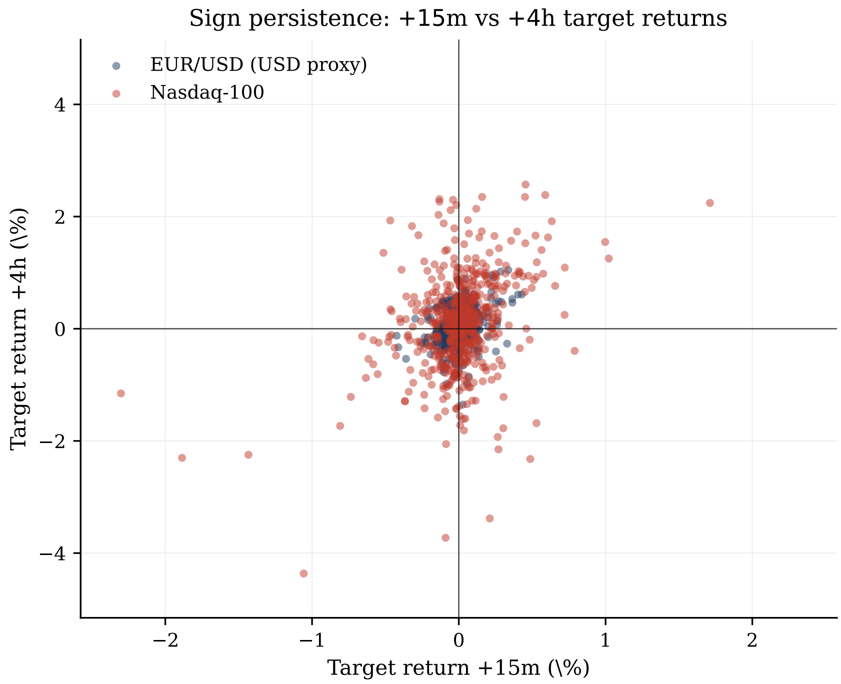

Figure 8: Target return at +15 minutes versus at +4 hours. Each point is one event-asset pair; colour distinguishes the two assets. Quadrants I and III correspond to sign agreement; the sample concentrates there at 57.6% of events.

6.5 Market structure

H4 (closed-period gap). When markets are closed (weekends, holidays), we aggregate the target sentiment of events occurring within the closed period and regress the realised opening gap on the aggregate sentiment. On NDX, the coefficient is +0.268 with $q = 0.021$ on 92 closed periods ($R^2 = 7.7\%$); on EUR/USD the coefficient is +0.102 with $q = 0.054$ on 32 periods ($R^2 = 10.3\%$). The NDX result is the cleaner of the two, consistent with the longer official non-trading window for equities than for FX.

H14 (cross-asset). The Pearson correlation between EUR/USD event-window returns and NDX event-window returns at 15 minutes is -0.148 ($q < 10^{-6}$), as expected from the macroeconomic risk-on/risk-off pattern under which USD strength tends to coincide with NDX weakness. Re-expressed in the USD-proxy convention, the correlation becomes +0.148 and the sign-match rate (USD-proxy and NDX moving in the same direction) is 56.6% across 1,176 clusters ($q < 10^{-5}$). We report both signs to forestall misinterpretation when the paper is read alongside event studies that work directly with the EUR/USD price.

6.6 Robustness

The Sections 6.1–6.5 results are supported by three robustness exercises.

C5 (pre/post 2026-01-15 knowledge-cutoff split). Splitting the sample at the LLM's training data cutoff, the central magnitude findings hold on both halves with $q < 10^{-13}$ throughout. On the pre-cutoff half, mean $|z|$ for the absolute return is 0.80–1.00 across primary windows on EUR/USD and 0.64–0.83 on NDX; the post-cutoff half is statistically similar. This argues against an explanation in which the LLM's effect is driven by memorisation of training-period news. Direction hit rates on the post-cutoff half remain within the same modest range reported in Section 6.2; the NDX 5-minute pre-cutoff hit rate is 55.8% with $q = 0.002$.

C6 (multivariate controls). A multivariate OLS of the standardised abnormal $|z|$ on category dummies, surprise level, expected magnitude, LLM confidence, log(cluster size), mean headline length, and pre-event move, with cluster-robust standard errors, recovers two controls that are consistently informative across cells: log(cluster size) ($q < 0.02$ in the EUR/USD 5-minute cell) and pre-event drift ($q < 0.002$ in the same cell). The cluster-size finding implies that larger news bursts produce larger reactions on average, even controlling for the sentiment and category labels. The pre-event-drift control complements the H8 result by showing that the event-window reaction is partially predictable from the run-up to the event. Category dummies absorb the energy and corporate effects identified in C4. Table 9 reports four representative cells.

Table 9: Multivariate regression of standardised abnormal magnitude.

	EUR/USD ($\lvert z\rvert$)		NDX ($\lvert z\rvert$)	
Term	+5m	+60m	+5m	+60m
log(cluster size)	1.925**	1.522***	1.057***	0.593*
	(0.751)	(0.413)	(0.367)	(0.323)
Pre-event $\lvert r\rvert_{15}$	8.516***	3.954***	3.529***	2.386**
	(2.480)	(1.337)	(0.954)	(0.956)
Surprise level	−0.010	−0.239	0.139	−0.228
	(0.259)	(0.194)	(0.140)	(0.167)
Expected magnitude	0.366	0.377*	−0.081	0.257
	(0.241)	(0.206)	(0.153)	(0.199)
LLM confidence	−0.031	−0.004	0.001	0.814
	(0.862)	(0.829)	(0.687)	(1.028)
Headline length	−0.001	0.001	−0.001*	−0.001*
	(0.001)	(0.001)	(0.000)	(0.000)
Cat: central bank	–	–	–	–
Cat: geopolitical	−0.744**	−0.393*	−0.104	−0.157
	(0.299)	(0.213)	(0.169)	(0.199)
Cat: corporate	−0.876*	−0.631*	−0.262	0.029
	(0.447)	(0.332)	(0.249)	(0.345)
Cat: energy	−0.952***	−0.124	−0.189	−0.399
	(0.290)	(0.316)	(0.190)	(0.248)
R^2	0.122	0.097	0.193	0.096
N clusters	1,279	1,202	1,188	1,088

Notes. OLS of the standardised abnormal $|z|$ on the listed regressors. Coefficient on top, robust standard error in parentheses below; cluster-robust covariance with the 15-minute news-cluster identifier as the grouping variable. $^*p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$ on raw p-values; q-values reported in the Online Appendix. Category baseline: `politics`.

C7 (winsorisation at the 1% tails). Winsorising every outcome at the 1% tails produces mean $|z|$ values that are within 5–10% of the raw mean and p-values that remain below 10^{-30} in the central cells. The headline magnitude finding is not driven by a small number of extreme observations. Table 11 reports the raw and winsorised means side by side.

Auxiliary LLM labels. The LLM's `expected_magnitude` and `surprise_level` labels (consolidating H5 and H13) have modest additional explanatory power, with ANOVA F-statistics that are significant at $q < 0.05$ on a minority of cells, primarily for NDX short

Table 10: Stability across LLM knowledge-cutoff split (15 January 2026).

Period	EUR/USD			NDX		
	5m	15m	60m	5m	15m	60m
Pre-cutoff: mean $\lvert z\rvert_{\lvert r\rvert}$	0.982	0.895	0.899	0.644	0.680	0.638
Post-cutoff: mean $\lvert z\rvert_{\lvert r\rvert}$	0.736	0.726	0.521	0.495	0.535	0.411
Pre-cutoff: q	1.9×10^{-15}	$<10^{-15}$	$<10^{-15}$	1.7×10^{-15}	7.8×10^{-15}	2.4×10^{-13}
Post-cutoff: q	1.7×10^{-10}	3.9×10^{-11}	2.3×10^{-8}	1.2×10^{-8}	3.1×10^{-11}	5.7×10^{-7}
Pre-cutoff N clusters	901	891	849	839	827	774

Notes. Standardised abnormal absolute return $|z|$ in the pre-cutoff and post-cutoff halves, with the split at the LLM's reported training data cutoff of 15 January 2026. q-values from one-sided t-test against zero with BH-FDR adjustment.

Table 11: Robustness to winsorisation at 1% tails.

Asset	Window	Mean $\lvert z\rvert$		q-value (t-test)		N
		Raw	Winsor 1%	Raw	Winsor 1%	
EUR/USD	+5m	0.908	0.790	$<10^{-15}$	$<10^{-15}$	1,286
EUR/USD	+15m	0.844	0.757	$<10^{-15}$	$<10^{-15}$	1,271
EUR/USD	+60m	0.787	0.709	$<10^{-15}$	$<10^{-15}$	1,207
NDX	+5m	0.599	0.549	$<10^{-15}$	$<10^{-15}$	1,198
NDX	+15m	0.637	0.589	$<10^{-15}$	$<10^{-15}$	1,176
NDX	+60m	0.571	0.527	$<10^{-15}$	$<10^{-15}$	1,097

Notes. Mean of $|z|$ (standardised abnormal absolute return) in the raw and 1%-winsorised samples, with corresponding q-values from a one-sided t-test against zero. Winsorisation is applied separately within each (asset, window) cell.

windows. We report the full table in the Online Appendix but recommend treating these labels as exploratory rather than as core inputs.

7 Discussion

The results in Section 6 are most usefully read as three separable themes: the magnitude–direction asymmetry of the news response, the methodological consequences of pre-event drift, and the limits of the present design relative to extensions we do not pursue.

7.1 Magnitude is easier to predict than direction

The result with the most uniform statistical signal across cells is that unscheduled news events are associated with substantially elevated intraday magnitude (Section 6.1), while the directional content of the same headlines is small and largely below transaction costs (Section 6.2). The two results are not in tension: they describe complementary properties of the same underlying phenomenon. A breaking headline produces price discovery—the market revises its mid-quote in response to new information—but the sign of the revision depends on context, prior positioning, and the fraction of the news that was already anticipated. Magnitude captures the price-discovery event itself; direction asks the harder question of which side of the new mid-quote the prior expectation was on.

From an applied perspective, news of this kind is more useful as a volatility filter than as a directional signal: volatility-aware position sizing, stop-loss design that allows for post-news range expansion, and execution rules that avoid mean-reversion trades immediately after a headline are natural operational uses of the magnitude finding. The directional content of the LLM sentiment, by contrast, is best treated as one weak predictor among many rather than as a standalone trigger. The pattern is consistent with Antweiler and Frank (2004), who find that message-board posting activity (a volume proxy) is more strongly associated with realised volatility than its bullishness content is with returns, and with the Heston and Sinha (2017) observation that news has faster effects on volatility than on directional returns.

The hit-rate advantage of the LLM-derived sentiment over both baselines is in the 2–6 percentage point range on the primary windows (Section 6.2), which is not a dramatic improvement on the headline metric. The qualitative advantage is more substantial: the LLM produces different sentiment for USD and for NDX on 73.5% of events, a property that captures the cross-asset asymmetry of risk-on/risk-off events that a single-polarity classifier cannot express. We recommend that subsequent work in this area report both metrics jointly, so that the cross-asset coherence of LLM labels with the known macro-finance taxonomy can be evaluated alongside the per-asset hit rate.

7.2 Pre-event drift: latency, not necessarily front-running

The pre-event drift result (Section 6.4, H8) is the methodologically most consequential finding of the analysis. The absolute return in the fifteen minutes preceding a Discord headline is elevated above the matched baseline at $q < 10^{-40}$ on both assets, with a magnitude comparable to the post-event drift on the same window.

A naive interpretation would invoke insider trading or systematic front-running. We caution against that reading. The Discord timestamp on which our event windows are aligned is downstream of the underlying primary source—typically a wire service, an agency feed, or an official release calendar—and the variable latency between the primary source and the Discord publication is, in our reading, the most plausible explanation. Without direct access to the primary-source timestamp for each event we cannot decompose the pre-event drift into "feed latency" and "true pre-disclosure activity"; we expect that the former dominates for most events.

The implication is the same regardless of the decomposition: the standard event-study assumption that the public publication time coincides with the informational event time is materially violated in this setting, and the apparent reaction in any post-event window includes a component that has, in fact, already taken place. Subsequent work on online-news event studies should either obtain primary-source timestamps or report the pre-event drift as a diagnostic alongside the post-event response.

7.3 What we do not test

We do not test whether more sophisticated LLM prompting (chain-of-thought, multi-prompt ensembles), more recent LLM models, or fine-tuning on a hand-labelled subsample would improve the directional results beyond the present 2–6 pp margin. We do not extend the asset universe beyond EUR/USD spot and the NDX CFD, nor do we test individual equities where firm-specific news effects would dominate. We do not decompose the magnitude response into a re-pricing component and a liquidity-withdrawal component, which would require an order-flow or quote-imbalance dataset we do not have. And we do not test cross-feed replication on a second public news service, which would quantify the feed-specific bias of the FinancialJuice gold filter. We view these extensions, particularly in combination with primary-source timestamps that would allow the pre-event drift to be decomposed cleanly, as the most promising directions for follow-up work.

8 Limitations

The analysis has five principal limitations. **First, single news feed and proxy timestamps.** We use one feed (FinancialJuice Discord) and its publication timestamp as the event time. The publication timestamp is downstream of the underlying primary source (wire service, agency feed, official release calendar), and the variable primary-source latency is, in our reading, the most plausible explanation for the pre-event drift documented in Section 6.4. The data alone cannot decompose this drift into "feed latency" and "true pre-disclosure activity"; obtaining primary-source timestamps for a sample of events, or cross-validating against a higher-tier feed (e.g., Bloomberg or Reuters Newscope), would resolve the issue.

Second, two-asset universe with a CFD proxy for the Nasdaq-100. The Dukascopy `E_NQ-100` series is a CFD price track for the Nasdaq-100, not the authoritative consolidated tape; we use it because it offers one-minute resolution with no gaps over our period. The findings should not be extrapolated to less-liquid currency pairs, to individual equities (where firm-specific news effects would dominate), or to fixed-income markets without explicit testing. The Dukascopy volume series for the CFD is tick volume rather than consolidated exchange volume; the H10 volume test is reported in the Online Appendix with this caveat explicit, and should not be interpreted as evidence on consolidated traded volume.

Third, the LLM labels are validated externally rather than on the event sample. Section 5.9 reports DeepSeek's 84.6% accuracy and $\kappa = 0.72$ on Financial PhraseBank (Malo et al. 2014), alongside relative comparisons with Loughran-McDonald and FinBERT on our sample. We have prepared a 200-event subsample for double-coded expert annotation in a follow-up study; the infrastructure is in place but the annotation is not yet complete. Additionally, the LLM-reported confidence field is uncalibrated as a probability of correctness (H6, reported here as a null finding) and is not used as a probability anywhere in the analysis. Two further null findings should be reported explicitly: H12 detects no robust bull-versus-bear asymmetry in the price response, and C8 (a cross-model consistency check on $n = 200$ events comparing `deepseek-v4-flash` and `deepseek-v4-pro`) is exploratory and underpowered for the inference suggested by the headline numbers.

Fourth, multiple testing and pre-specification. The pipeline runs more than one hundred individual p-values across the main tests, the auxiliary tests, and the cells of stratified analyses. We address this with uniform Benjamini-Hochberg adjustment (Benjamini and Hochberg 1995) and draw primary conclusions exclusively from q-values. A residual concern is that the C1–C7 extensions were added during methodological revision after the H1–H14 pre-specification; the C extensions are clearly demarcated in the codebase and a stricter pre-registration standard can be applied by re-reading Section 6 attending only to the H1–H14 results. The central conclusions of Sections 6.1 (H1) and 6.4 (H8 and H9) survive this restriction.

Fifth, dependence structure. Cluster-dependence at sub-15-minute frequencies means adjacent events in a news burst are not independent. We address this with cluster-dedupe in non-regression tests and cluster-robust standard errors in regression tests; both devices may under-correct in long news bursts (cluster sizes occasionally exceed twenty events). Suggested extensions for follow-up work, in priority order, are: (i) the manual validation and primary-source-timestamp work mentioned above; (ii) cross-feed replication on a second public news service; (iii) cross-asset extension to a basket of individual equities for firm-specific news; and (iv) prompt engineering and LLM ensembling on the directional task to test whether the modest direction edge can be improved beyond the present 2–6 pp margin over baselines.

9 Conclusion

We document the intraday price response of EUR/USD and the Nasdaq-100 to a thirteen-month sample of 2,449 unscheduled news events from a public real-time newsfeed, with each event independently labelled by a large language model and benchmarked against the standard Loughran-McDonald dictionary and the FinBERT-tone neural classifier. After uniform Benjamini-Hochberg false discovery rate adjustment, the news events are robustly associated with intraday magnitude that is 1.3× to 1.9× the matched-baseline level on every asset-window cell tested, and this finding is stable under winsorisation, multivariate controls, and a pre/post-knowledge-cutoff split. The directional content of the LLM sentiment is modest (49–54% hit rate against realised direction) and below transaction costs in a backtest, but the LLM nonetheless outperforms the Loughran-McDonald baseline by 2–6 percentage points and the FinBERT baseline by up to 4.5 percentage points on the primary windows. The LLM is also the only one of the three classifiers capable of producing asset-specific sentiment (different USD and NDX labels on the same headline), which it does on 73.5% of events.

Two methodologically substantive findings sit alongside the main results. The pre-event drift in the fifteen minutes before each Discord headline is comparable in magnitude to the post-event drift, suggesting that the public publication time is not the informational event time in this setting; we recommend that future work on online-news event studies obtain primary-source timestamps where possible and report the pre-event drift as a diagnostic in any case. And the initial price reaction extends rather than reverses over the subsequent hours, with the sign of the +15-minute return agreeing with the sign of the +4-hour return on 57.6% of events and median absolute moves 3–4.5× larger over the longer window.

Two implications follow. For empirical work in this area, an LLM-based sentiment classifier provides meaningful value over the standard dictionary and BERT-family baselines, both in hit rate and in the asset-specific expressiveness that single-output classifiers cannot match. For the operational interpretation of the news-flow, the magnitude signal is more reliable than the directional signal: news events are most defensibly understood as triggers for volatility expansion, with direction available only as a weak secondary input that does not overcome execution costs in the naive specifications tested here.

We make the full pipeline—including the validation harness, the LLM sentiment cache, the dictionary and FinBERT baselines, and the three-way comparison—publicly available in a single repository with seeded reproducibility, in the hope that the methodology can be re-applied to other public newsfeeds and to a wider set of assets.

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Data Availability Statement

The replication package supporting this study—comprising the full analysis pipeline, the LLM sentiment cache, the Loughran-McDonald and FinBERT baseline runs, the external benchmark replication, the table and figure generation scripts, and the LaTeX source for this paper—is available at [GITHUB_REPO_URL, TBA].

The two raw input datasets are publicly accessible and reproducible from their primary sources rather than redistributed in the replication package:

- **News data.** FinancialJuice Discord newsfeed, channel `1353792641392971776`. Export performed with DiscordChatExporter (open source). The 63,016-message JSON archive is reproducible from a fresh export over the same date range (24 March 2025 to 5 May 2026, UTC). Access requires Discord authentication; the channel is publicly readable to verified accounts.
- **Price data.** One-minute OHLCV bars for EUR/USD spot (Dukascopy) and the Nasdaq-100 CFD (`E_NQ-100` on Dukascopy). Reproducible via the `dukascopy-python` command included in the replication package (`download_prices.py --start 2025-03-24 --merge-existing`). No Dukascopy account is required.
- **External benchmark.** Financial PhraseBank (Malo et al. 2014), accessed via Hugging Face Datasets at `lmassaron/FinancialPhraseBank`.

The DeepSeek API sentiment cache is shipped with the replication package as a SQLite database, so re-running the LLM classification step is not required to reproduce the reported results. Re-running from scratch with the published prompt and seed reproduces the cache deterministically modulo API non-determinism in model responses (temperature is set to zero throughout).

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A Reproducibility Statement

All numerical results in this paper can be reproduced from the public project repository using the commands and the seed reported below.

Code, data, and environment

The repository is organised around eight pipeline scripts plus auxiliary tools. Production runs use Python 3.13 in an isolated virtual environment with the dependencies pinned in `requirements.txt`. The event-study core uses `pandas`, `numpy`, `scipy`, `statsmodels`, and `matplotlib`; the LLM sentiment step uses the `openai` Python client against a DeepSeek API endpoint; and the dictionary baseline uses `pysentiment2`, which bundles the Loughran-McDonald master dictionary (Loughran and McDonald 2011). The FinBERT baseline uses `transformers` (4.x) with the `yiyanghkust/finbert-tone` pre-trained model (Yang et al. 2020).

Raw price data are downloaded via `dukascopy-python` (one-minute OHLCV bars for EUR/USD and the `E_NQ-100` CFD). Raw news data are exported via DiscordChatExporter from the publicly accessible FinancialJuice newsfeed channel and stored as a single JSON archive under `data/`. Neither input file is committed to the repository; both are reproducible from the underlying public sources.

Random seeds

All stochastic procedures (matched baseline sampling, bootstrap resampling, train/test splits used in robustness checks) are seeded with `SEED = 42` in `event_study.py`. Rerunning the pipeline with the same code, same input data, and the same seed reproduces every numerical value reported in Section 6 to the full precision of the output CSVs.

End-to-end reproduction

The full pipeline is reproduced by the following sequence of commands. Steps 1–3 download or generate the inputs; steps 4–9 produce the analysis outputs.

```
# 1. Parse the Discord export into the gold event list.
python parse_fj_discord.py "data/FinancialJuice ... .json" \
    -o outputs/events.csv --summary

# 2. Download or refresh the 1-minute price panels.
python download_prices.py --start 2025-03-24 --merge-existing

# 3. Classify each gold event with the LLM (cached in SQLite).
python sentiment.py outputs/events.csv \
    -o outputs/events_sentiment.csv --workers 15

# 4. Run the event-study pipeline (H1-H14, C1-C7).
python event_study.py

# 5. Validate methodology-sensitive outputs.
python validate_outputs.py
# Expected output: "OK: outputs passed methodology sanity checks"

# 6. Generate the HTML report.
python make_report.py --also-copy docs/report.html

# 7. Run the Loughran-McDonald dictionary baseline.
python sentiment_baseline.py outputs/events_sentiment.csv \
    -o outputs/events_sentiment_baseline.csv

# 8. Run the FinBERT-tone neural baseline (first run downloads
#    the yiyanghkust/finbert-tone model, ~440 MB, into HF cache).
python sentiment_finbert.py outputs/events_sentiment.csv \
```

```
    -o outputs/events_sentiment_finbert.csv

# 9. Three-way comparison on event sample: LLM vs LM vs FinBERT.
python sentiment_baseline_compare.py

# 10. External benchmark validation on Financial PhraseBank
#     (cached in outputs/sentiment_external_cache.sqlite).
python sentiment_external_benchmark.py
```

The manual-validation subsample for the sentiment audit is generated by `prepare_manual_validation.py` and scored, once the two-annotator labels are filled in, by `score_manual_validation.py`.

Validation harness

`validate_outputs.py` runs deterministic sanity checks on the generated outputs: it verifies that the event/baseline magnitude ratio in `h1_results.csv` is between 1.0 and 5.0; that the H2 hit-rate column is in $[0, 1]$; that `methodology_summary.csv` reports the expected target conventions; and that the cluster-level outputs are non-empty. A pipeline run with code changes that breaks any of these invariants will fail this check before any results are reported.

Pre-specification

The fourteen pre-specified hypotheses H1–H14 are defined in `event_study.py` and were fixed before any cleanup of the result tables. The extensions C1–C7 were added during methodological revision, as documented in `STATUS.md`; C8 was added as an exploratory analysis and is reported in Section 8 rather than as a main result. Robust standard errors, cluster-robust standard errors, and the Benjamini-Hochberg false discovery rate adjustment (Benjamini and Hochberg 1995) are applied uniformly to every test output and are not chosen post hoc per hypothesis.

Repository

The complete source for the analysis and this paper draft is maintained as a git repository. Each subsection in Section 6 corresponds to a CSV file in `outputs/`; readers seeking to audit a specific number reported in the text can locate the underlying row by the variable name conventions described in `validate_outputs.py`.

B Online Appendix: Robustness Tables

This appendix collects auxiliary tables referenced in the main body. All quantities are computed from the same pipeline and seed documented in Section A.

B.1 Cluster-gap-threshold robustness for H1

Section 5.3 treats the 15-minute cluster-gap threshold as a design parameter and notes that the central H1 result is insensitive to the choice within the range 5–30 minutes. Table 12 reports the H1 event/baseline magnitude ratio for the four thresholds. The ratio is monotone in the threshold (tighter thresholds retain more independent clusters and yield slightly higher ratios), but remains above 1.0 in every cell, the central conclusion in Section 6.1.

Table 12: Cluster-gap-threshold robustness for the H1 magnitude ratio.

		Cluster gap threshold			
Asset	Window	5 min	10 min	15 min	30 min
EUR/USD	+5m	1.90×	1.85×	1.76×	1.67×
EUR/USD	+15m	1.83×	1.77×	1.69×	1.60×
EUR/USD	+60m	1.66×	1.62×	1.58×	1.50×
NDX	+5m	1.73×	1.65×	1.56×	1.48×
NDX	+15m	1.71×	1.66×	1.57×	1.47×
NDX	+60m	1.57×	1.51×	1.43×	1.38×
Number of clusters after dedupe					
EUR/USD +5m		1,557	1,395	1,288	1,069
EUR/USD +15m		1,539	1,379	1,273	1,057
EUR/USD +60m		1,469	1,313	1,208	1,001
NDX +5m		1,457	1,305	1,200	993
NDX +15m		1,435	1,283	1,178	972
NDX +60m		1,350	1,200	1,099	910

Notes. H1 event/baseline magnitude ratio at four cluster-gap thresholds. The 15-minute column corresponds to the specification used in the main body. Tighter thresholds (5–10 min) retain more independent clusters and yield slightly higher ratios; a wider threshold (30 min) compresses the sample but the central finding survives in every cell.

B.2 Subsample analysis by LLM sentiment label

The bull-versus-bear asymmetry test (H12) reported as null in Section 8 aggregates over all events. Tables 13 and 14 report the magnitude and direction results stratified by the LLM sentiment label (bull, bear, neutral). The magnitude effect is broadly symmetric across the three subsamples and confirms the H12 null finding at a more granular level. The direction hit rate by sentiment label shows one notable cell: EUR/USD events labelled bull achieve a 57.3% hit rate at the +60-minute window ($p = 0.0007$ on a one-sided binomial); the corresponding bear subsample is not statistically distinguishable from chance. We report this observation as exploratory; it does not feature in the main results because the same conclusion would not hold under a stricter joint-testing correction across the eighteen subsample cells reported here.

Table 13: Magnitude by sentiment label (subsample analysis).

Asset	Window	Sentiment	N	Mean $\|r\|$ (%)	Median $\|r\|$ (%)
EUR/USD	+5m	bull	557	0.0333	0.0207
EUR/USD	+5m	bear	464	0.0330	0.0207
EUR/USD	+5m	neutral	384	0.0306	0.0188
EUR/USD	+15m	bull	557	0.0547	0.0346
EUR/USD	+15m	bear	464	0.0527	0.0360
EUR/USD	+15m	neutral	384	0.0518	0.0311
EUR/USD	+60m	bull	557	0.1006	0.0677
EUR/USD	+60m	bear	464	0.1025	0.0714
EUR/USD	+60m	neutral	384	0.0935	0.0630
NDX	+5m	bull	475	0.0816	0.0506
NDX	+5m	bear	649	0.0923	0.0554
NDX	+5m	neutral	281	0.0700	0.0458
NDX	+15m	bull	475	0.1409	0.0933
NDX	+15m	bear	649	0.1471	0.0931
NDX	+15m	neutral	281	0.1430	0.0866
NDX	+60m	bull	475	0.2619	0.1606
NDX	+60m	bear	649	0.2806	0.1745
NDX	+60m	neutral	281	0.2539	0.1751

Notes. Event-window absolute return $|r|$ stratified by LLM sentiment label. The magnitude effect is broadly symmetric across bull, bear, and neutral subsamples; the small differences do not motivate a separate H12-style asymmetry claim.

Table 14: Direction hit rate by LLM sentiment label.

Asset	Window	Sentiment	N	Hit rate	CI95	p-binom
EUR/USD	+5m	bull	523	50.1%	[0.458, 0.544]	0.500
EUR/USD	+5m	bear	475	54.9%**	[0.505, 0.594]	0.017
EUR/USD	+15m	bull	521	52.4%	[0.481, 0.567]	0.147
EUR/USD	+15m	bear	470	52.6%	[0.480, 0.570]	0.144
EUR/USD	+60m	bull	494	57.3%***	[0.529, 0.616]	0.0007
EUR/USD	+60m	bear	450	50.4%	[0.458, 0.550]	0.444
NDX	+5m	bull	441	54.0%*	[0.493, 0.586]	0.053
NDX	+5m	bear	560	51.8%	[0.476, 0.559]	0.211
NDX	+15m	bull	434	53.5%*	[0.488, 0.581]	0.082
NDX	+15m	bear	551	49.7%	[0.456, 0.539]	0.568
NDX	+60m	bull	403	53.8%*	[0.490, 0.587]	0.067
NDX	+60m	bear	515	46.2%	[0.420, 0.505]	0.961

Notes. Direction hit rate separately for events labelled bull and bear by the LLM (neutral events excluded as non-directional). Bull-labelled EUR/USD events at +60 minutes reach 57.3% hit rate ($p = 0.0007$). $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$ on raw p-values.

B.3 LLM prompt full text

The production LLM call uses the system prompt below. The benchmark-adapted prompt for Financial PhraseBank (Section 5.9) is identical to this prompt up to the schema (positive/negative/neutral in place of the USD/NDX bull/bear/neutral fields) and the omission of the asset- specific instruction.

```
You are a financial sentiment analyst. Classify how a news event
likely affects the US Dollar (USD) and the Nasdaq-100 index (NDX) in
the very short term (minutes to hours after publication). Output
STRICTLY a single JSON object with this exact schema: sentiment_usd
∈ {bull, bear, neutral}, sentiment_ndx in the same support, signed
directional_strength_usd and directional_strength_ndx in [−1, +1],
expected_magnitude in {low, med, high}, surprise_level in {expected,
surprise, shock}, confidence in [0, 1], and a short rationale.

Rules: base judgment strictly on the supplied text; do not invent
context; if ambiguous or off-topic, return neutral with low confidence
(<0.4); USD and NDX are different assets and their sentiments can
differ (e.g., risk-off events: bull USD, bear NDX); directional_strength
values respect sentiment direction; return ONLY the JSON object, no
preamble.
```

Five few-shot examples covering geopolitical escalation, dovish Fed signal, hawkish ECB signal, neutral aid announcement, and corporate earnings miss anchor the format.